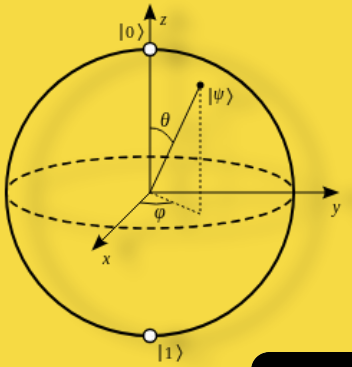
Quest for QML Advantage

In Medicine



Bryan Cardenas
High Performance Machine Learning

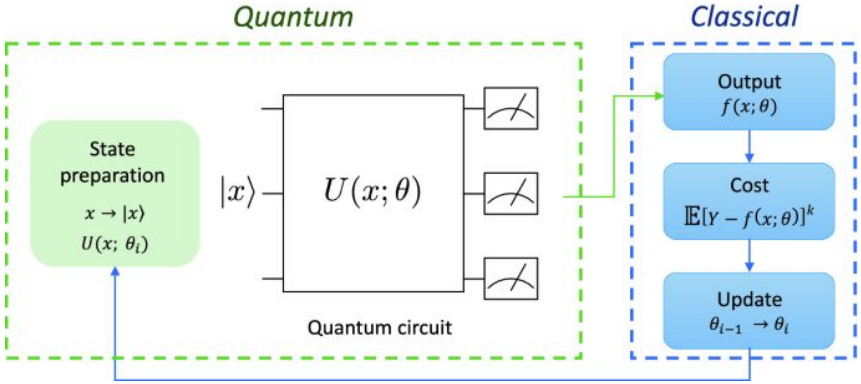
SURF

NISQ-Era

1. Noisy Intermediate-Scale Quantum is defined by hardware constraints
   a. Limited qubits
   b. High Noise & Decoherence

2. Stochasticity
   a. Non-computational Basis State (frequencies of outcomes)
   b. Trade-off between precise measurement of single state vs the less precise many

3. Employ a Hybrid Quantum-Classical Algorithmic Approach
   a. Use Variational Quantum Algorithms



CANNOT RELY ON ANY TYPE OF ERROR CORRECTION

SURF

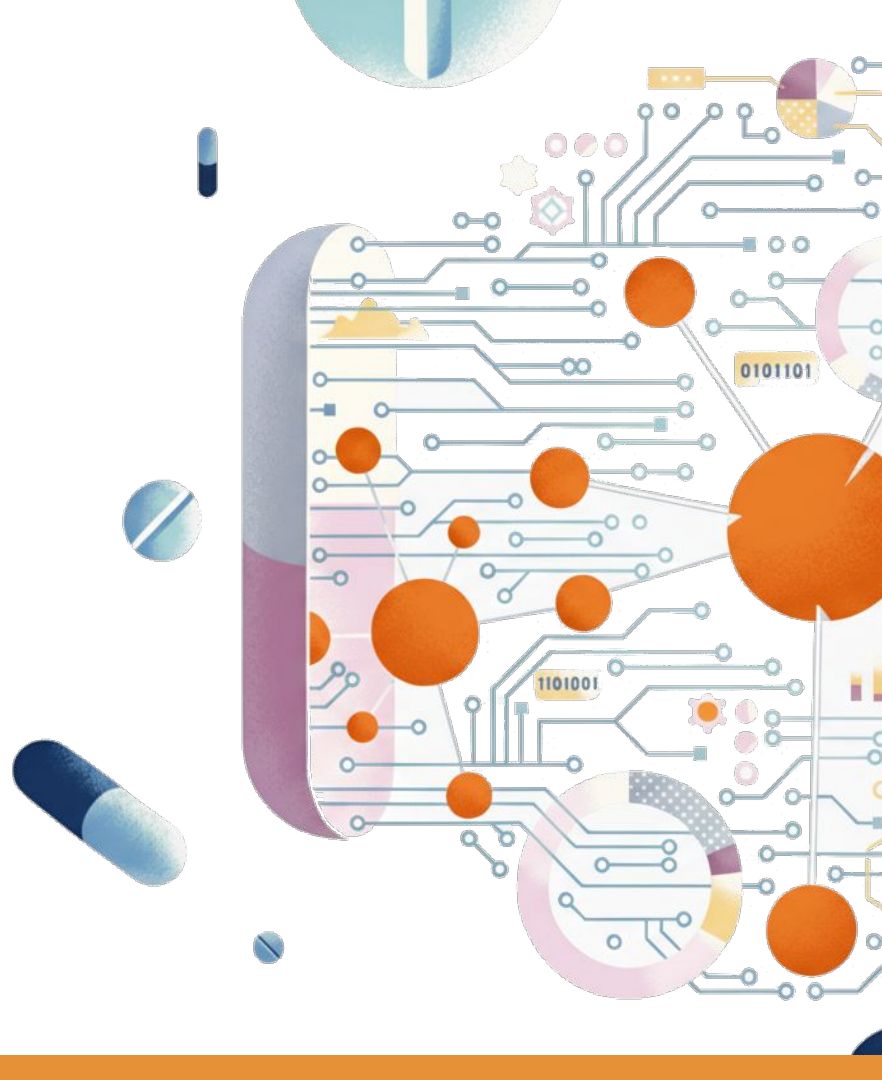
The Quest

1. The Goal

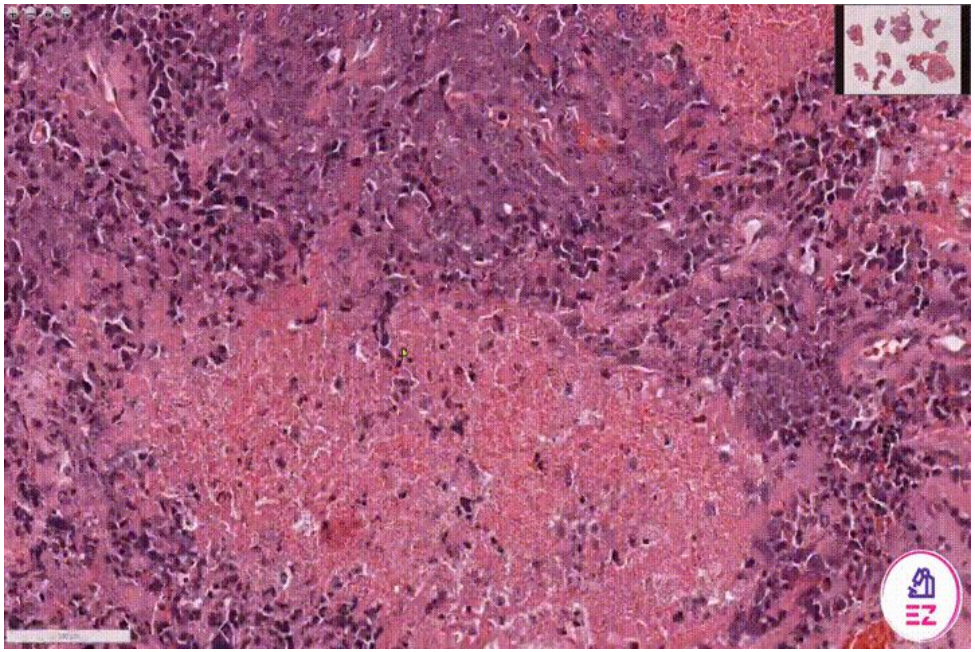
Find where quantum advantage lives in the intersection QML and Medicine

2. Strategy

Employ a hybrid classical-quantum model for feature extraction; see in what ways we have an advantage over CML



Histopathology Example

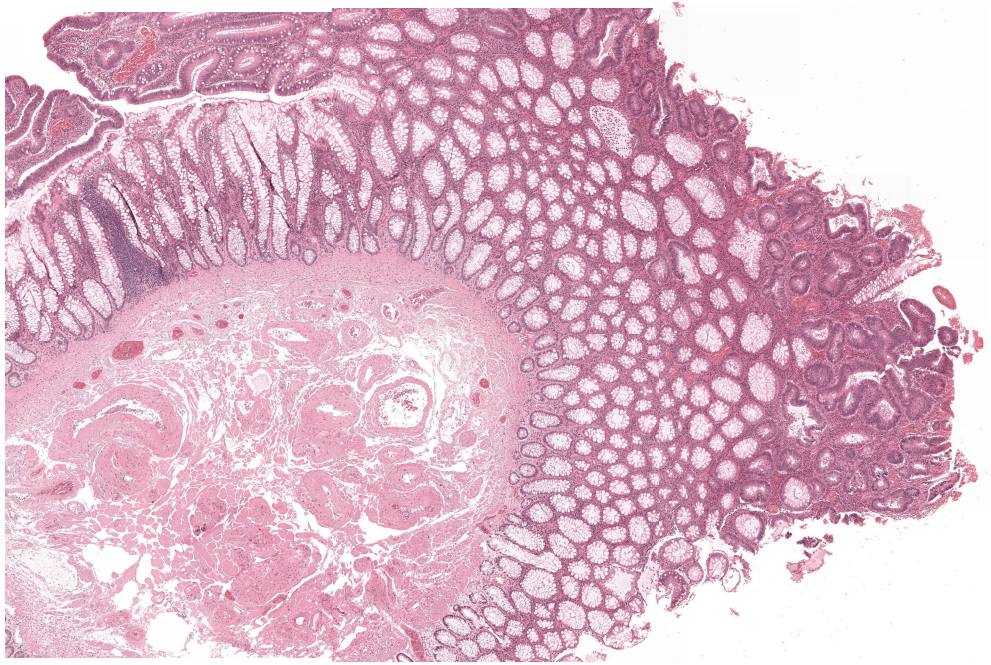


1. **Data Dimensionality**
   Input can be extremely large

2. **Complex Features**
   A lot of (local) features remain *unseen* for the pathologist

   Can a VQC's access to high-dimensional Hilbert Space lead to more more complex decision boundaries?

Histopathology Example

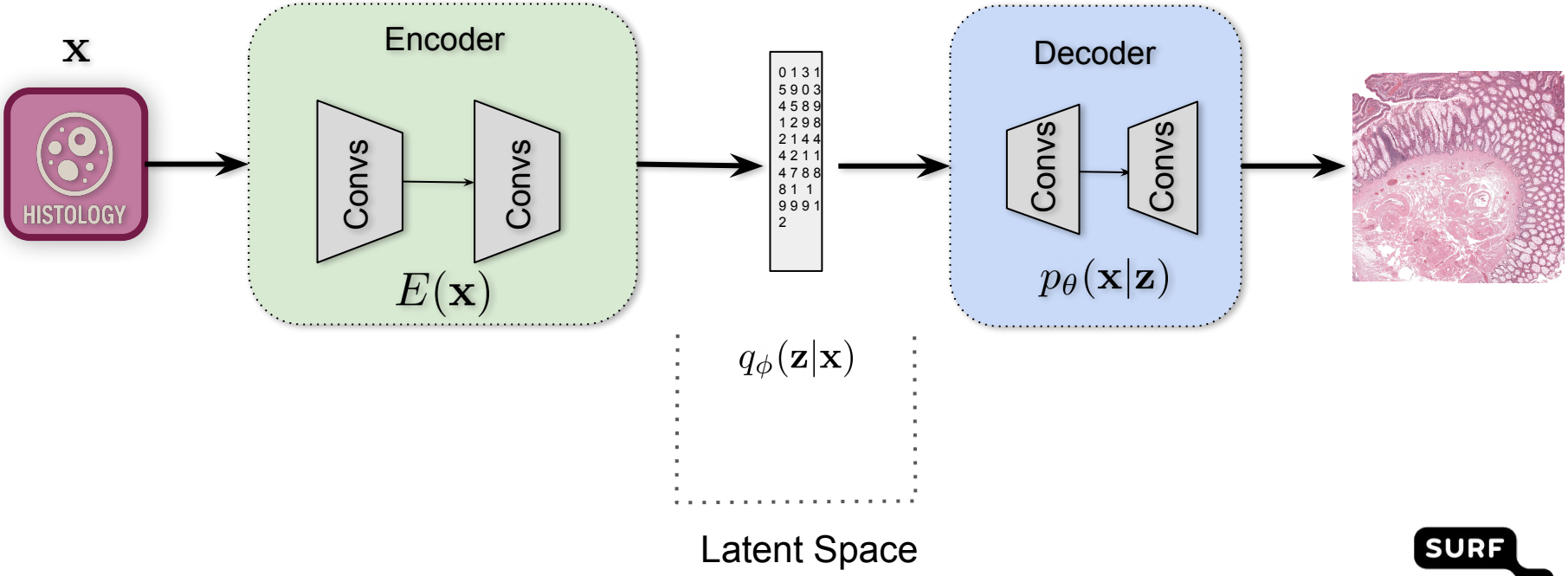


1. **Data Dimensionality**
   Input can be extremely large

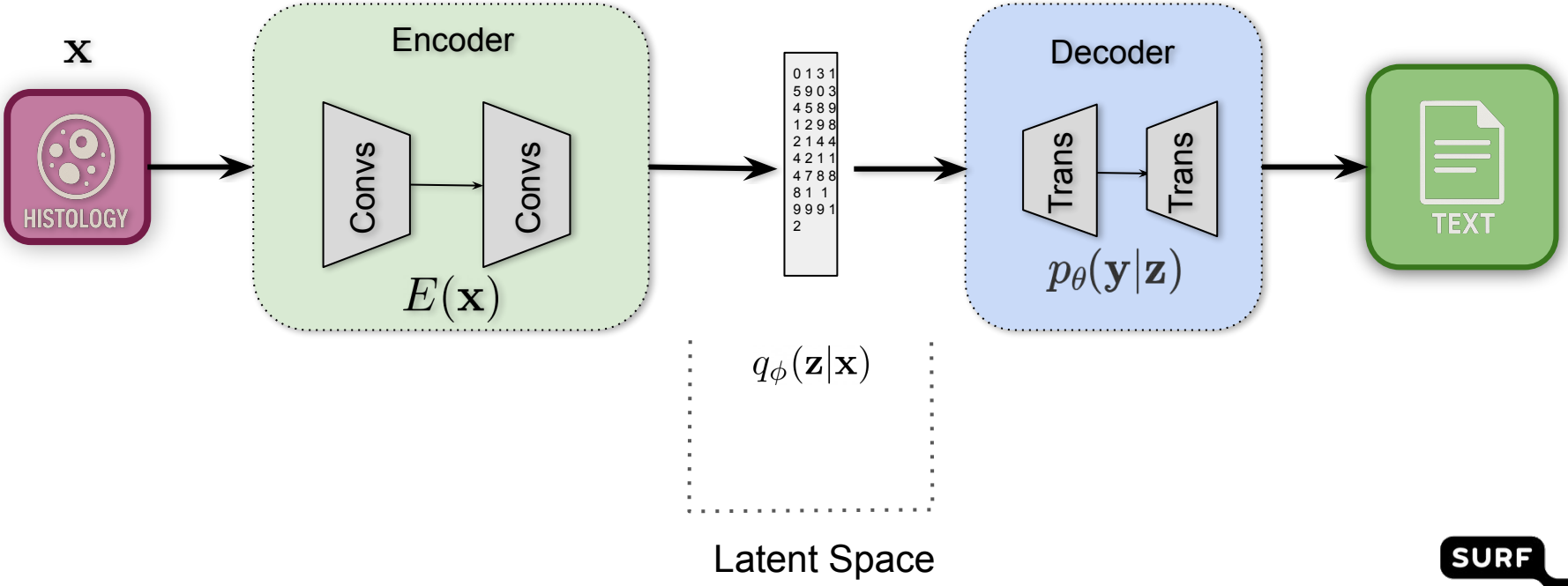
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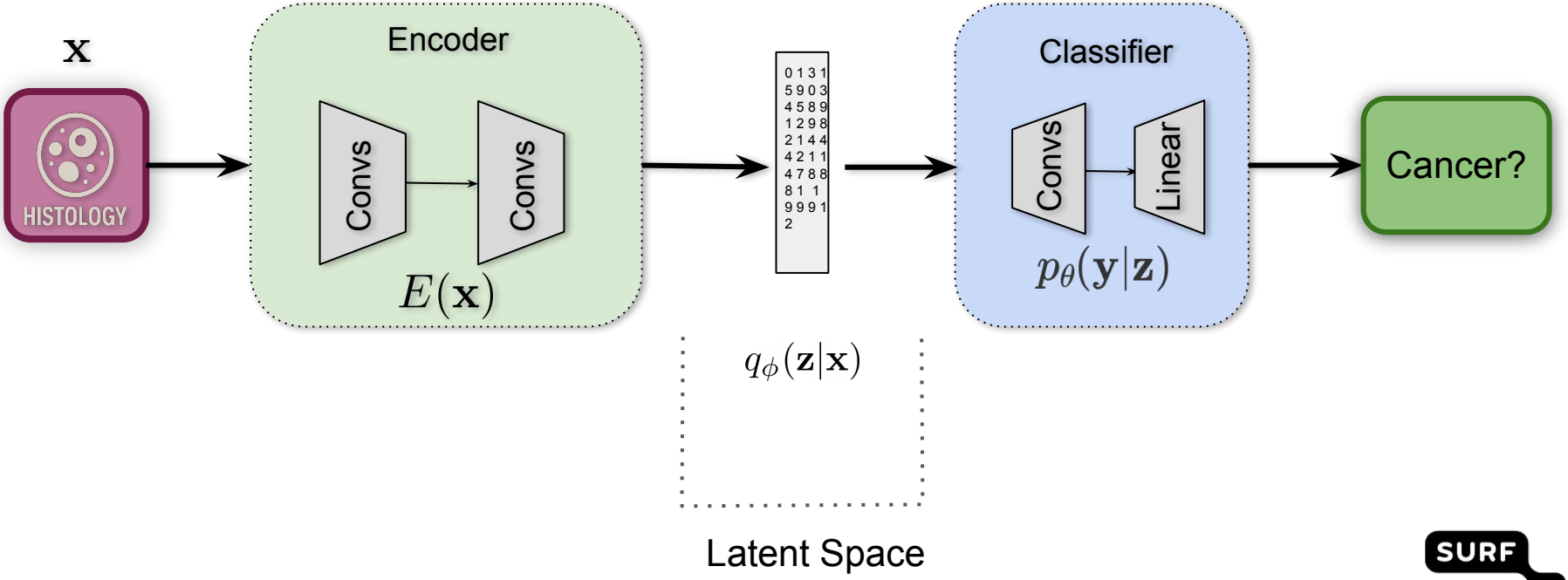
Histopathology Classical ML



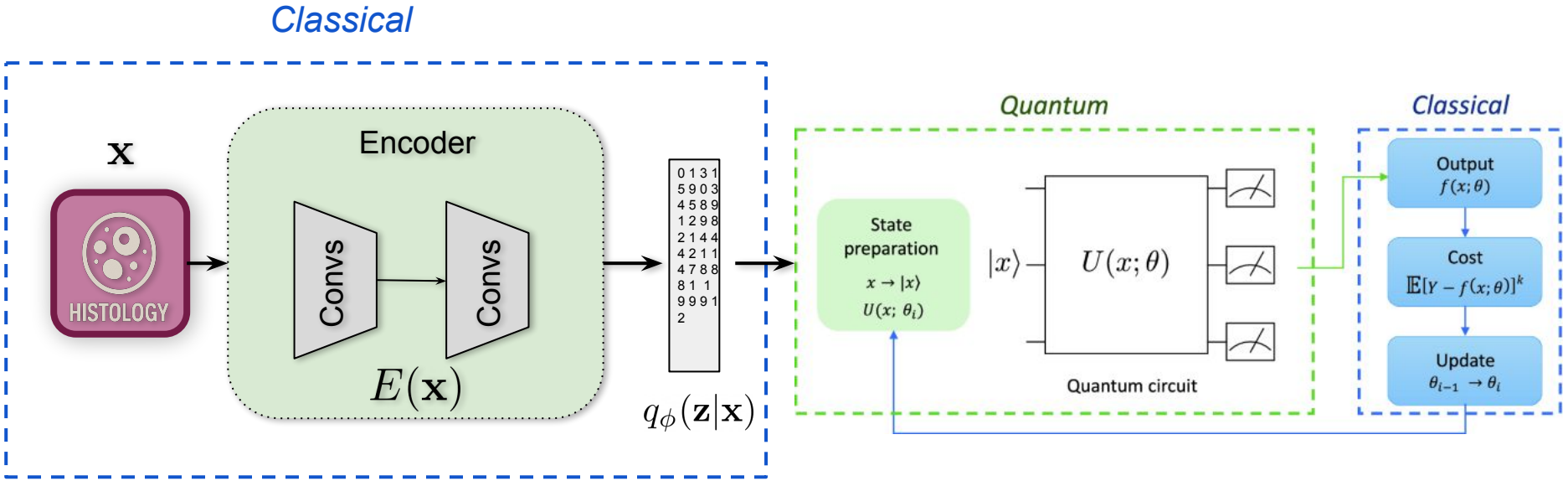
Histopathology Classical ML



Histopathology Classical ML

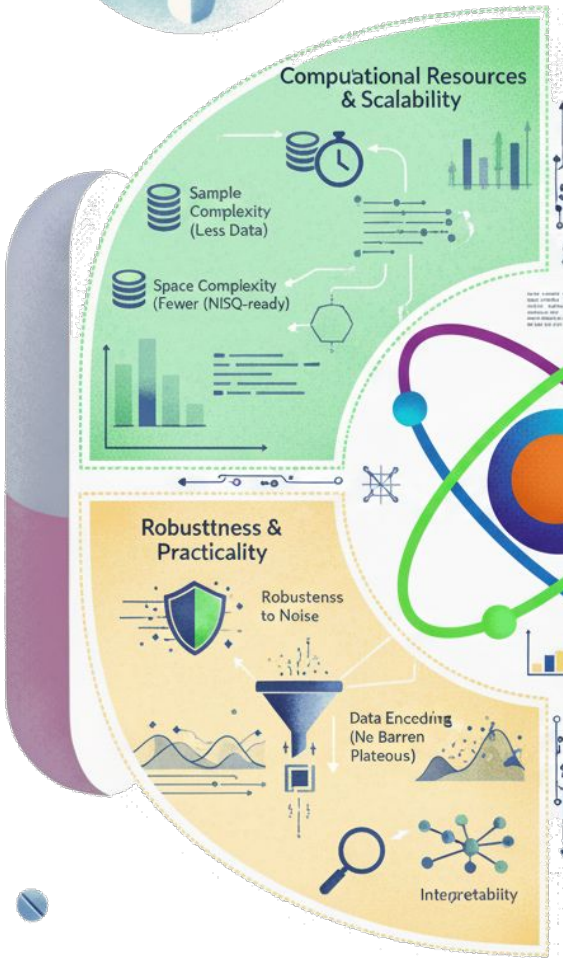


Histopathology Hybrid QML Model



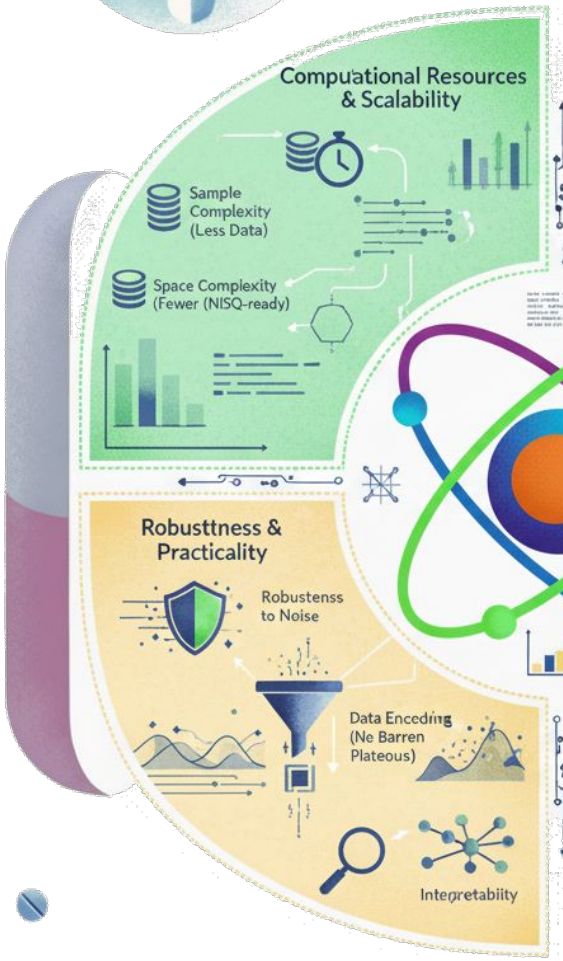
The Breadcrumbs

1. Convergence Speed
   a. Sample complexity
2. Time/Space Complexity
3. Model Expressivity (Accuracy)
   a. Latent Space Entanglement
4. Data Encoding Efficiency
5. Trainability



The Breadcrumbs

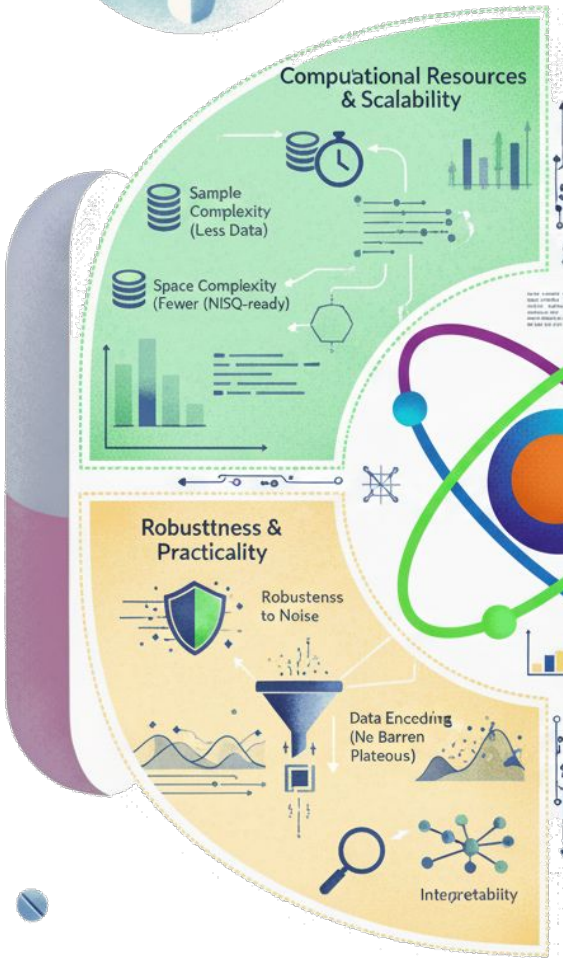
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The Breadcrumbs

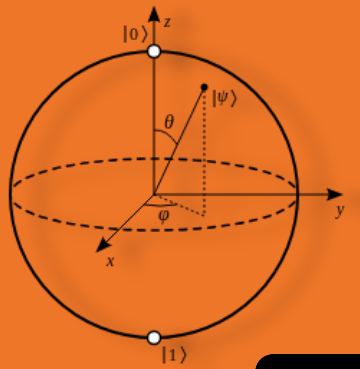
1. Convergence Speed
   a. Sample comple
2. Time/Space Complexity
3. Model Expressivity (cy)
   a. Latent Space E glement
4. Data Encoding Efficiency
5. Trainability

Fault Tolerant QC



SURF

Data and QML

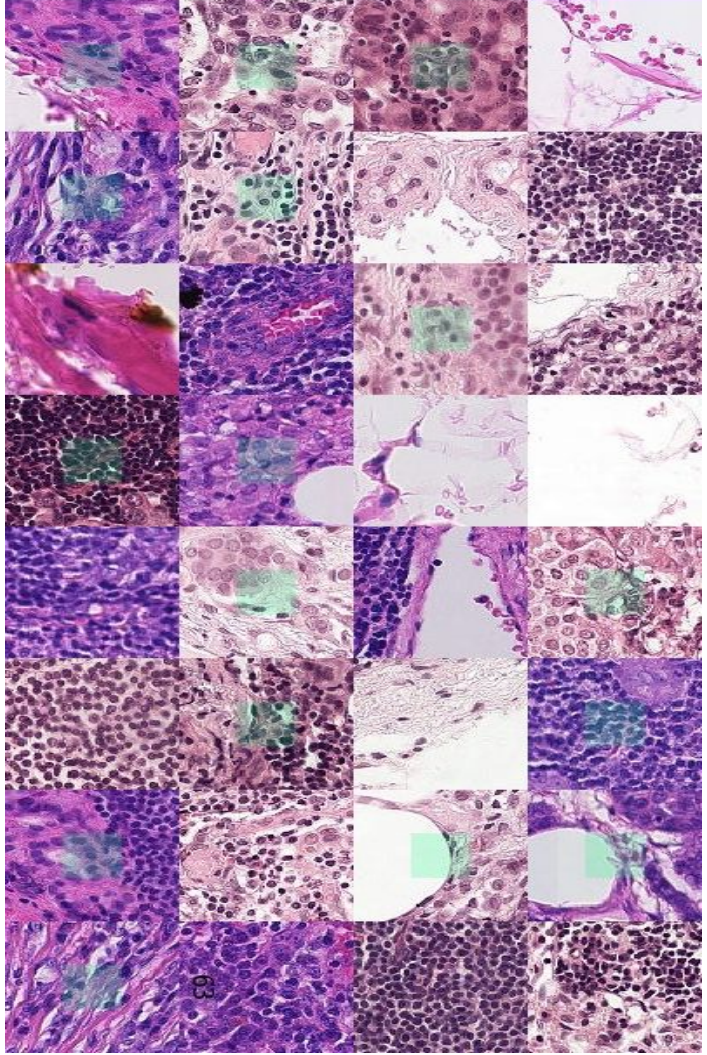


SURF

High Performance Machine Learning Group

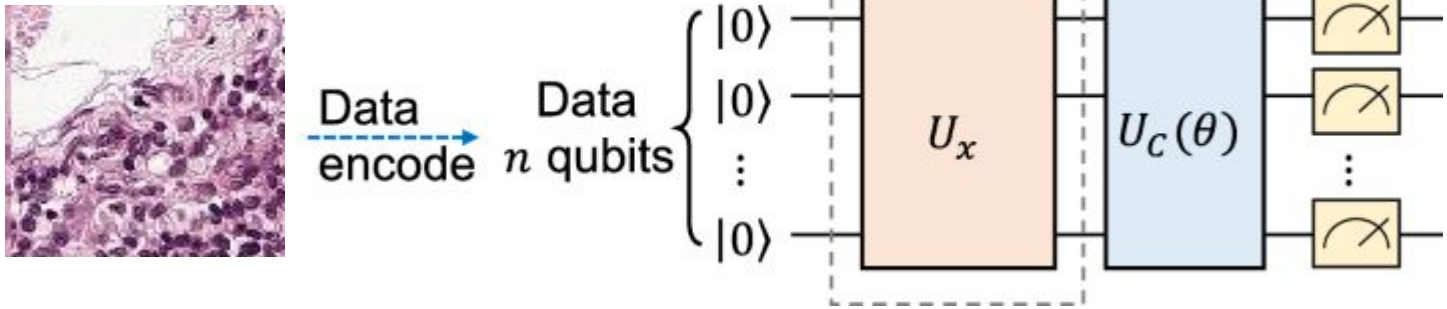
PatchCamelyon (PCam)

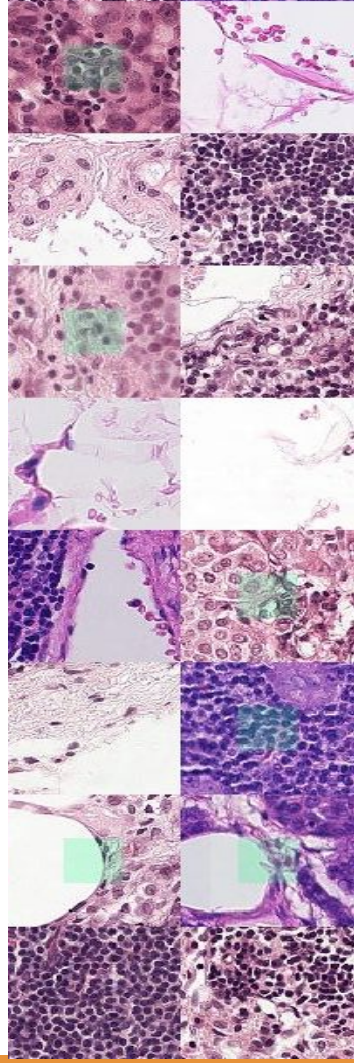
1. Not too easy, like MNIST, sufficiently complex
2. 96x96x3 dimensional images
3. Binary Labelled (metastatic tissue)



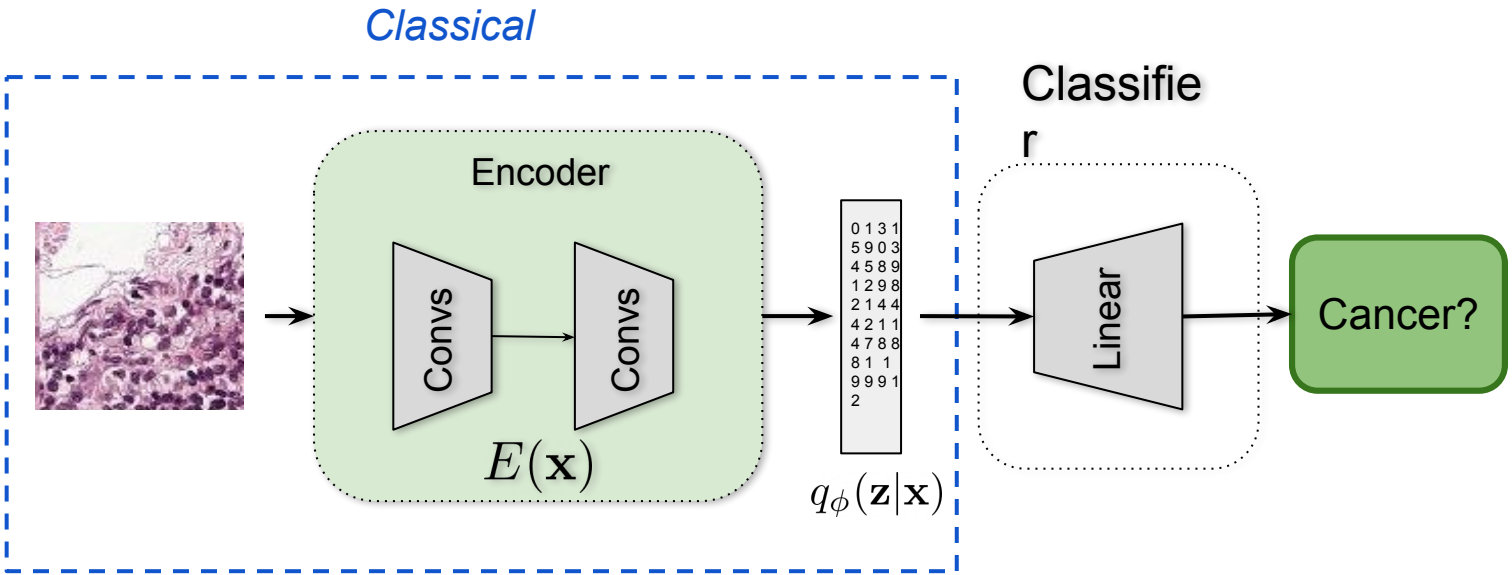
https://github.com/basveeling/pcam?tab=readme-ov-file

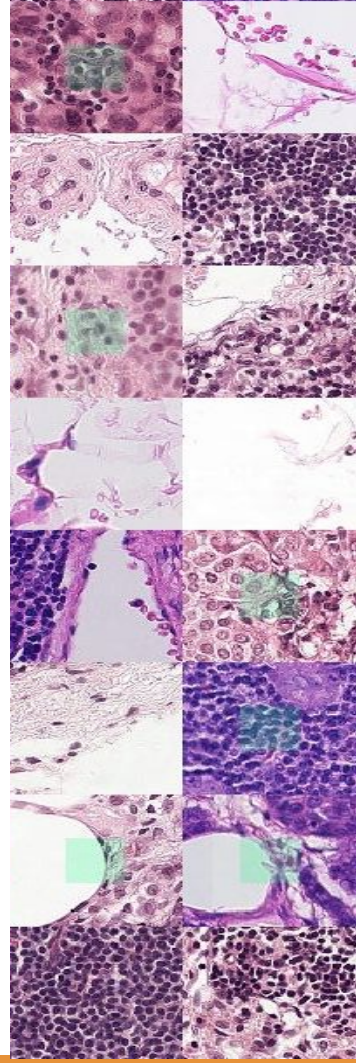
PatchCamelyon (PCam)



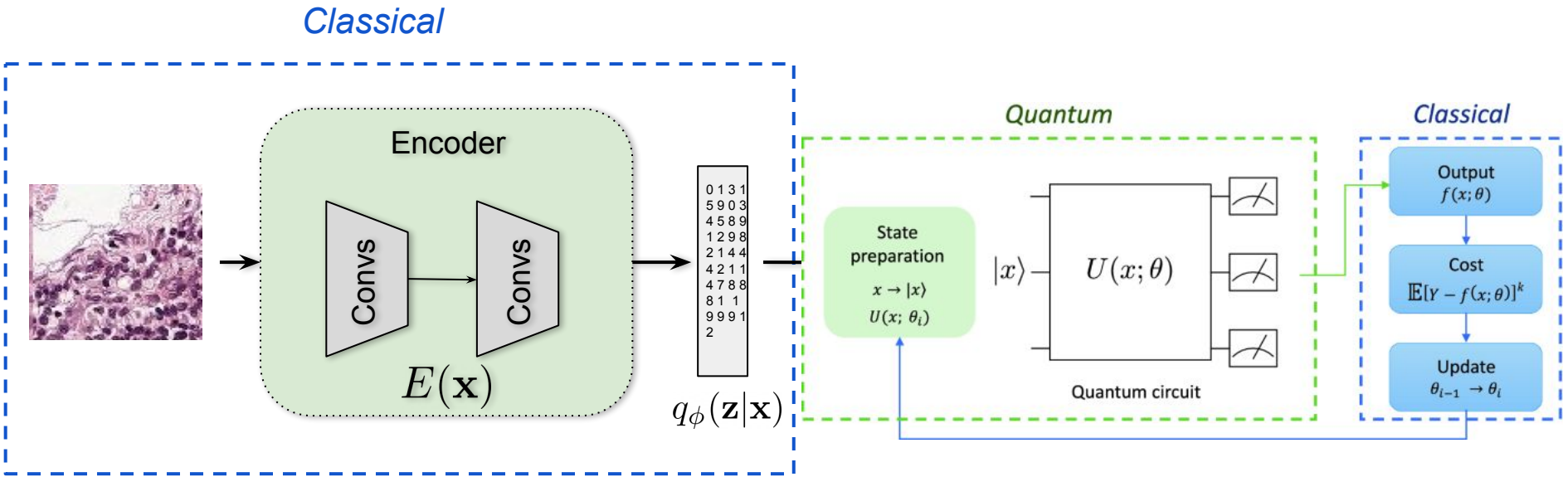


Classical Classifier

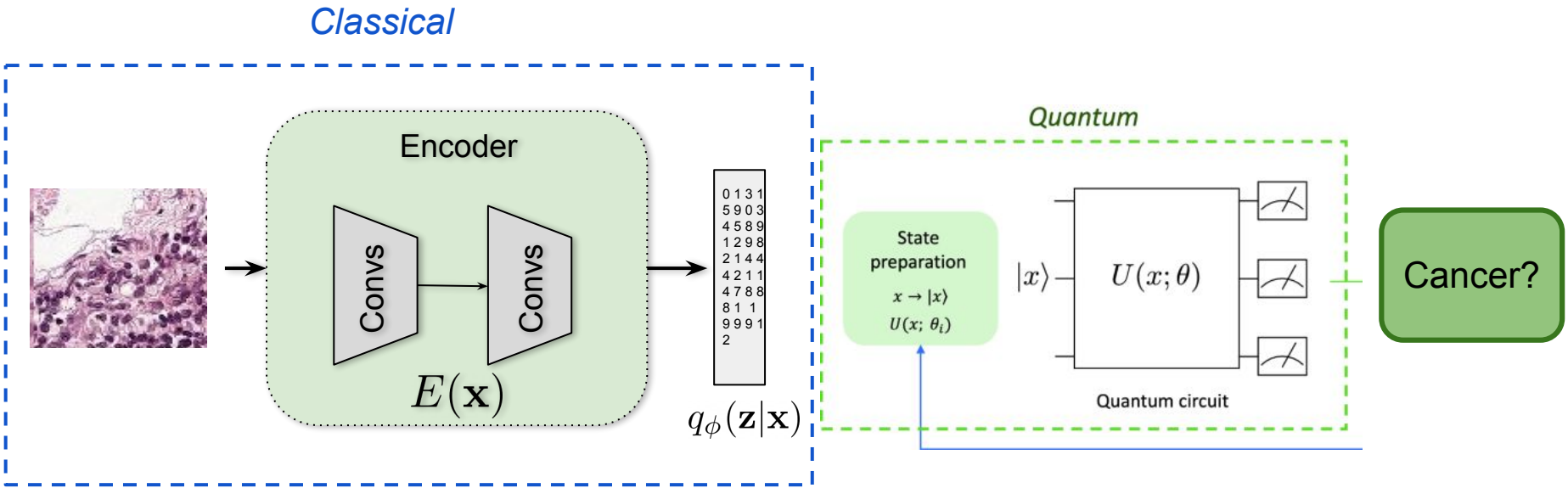




Hybrid Classifier

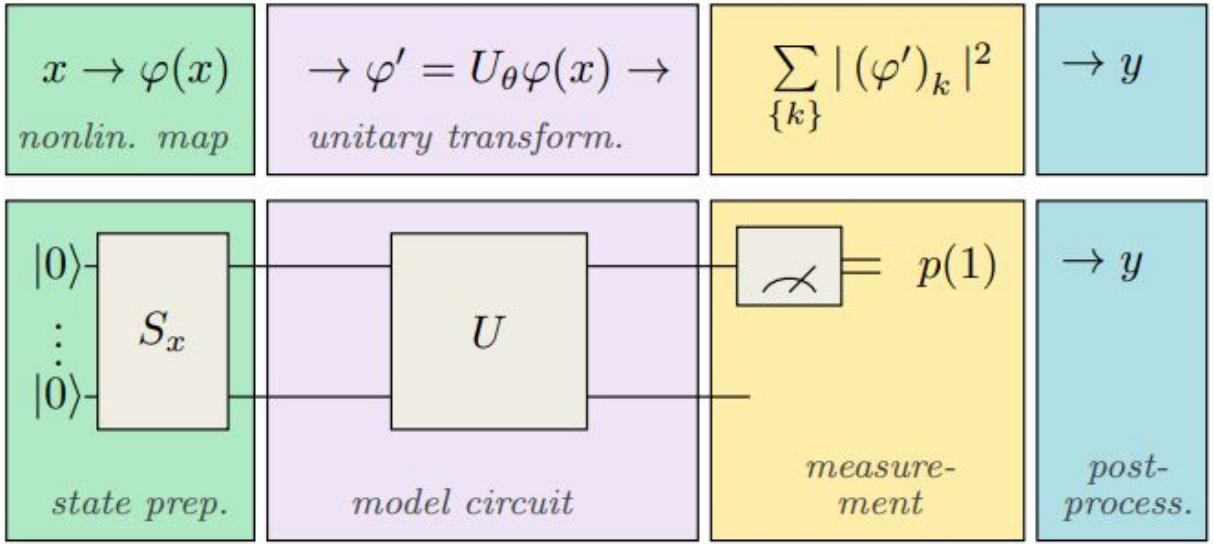


Hybrid Classifier

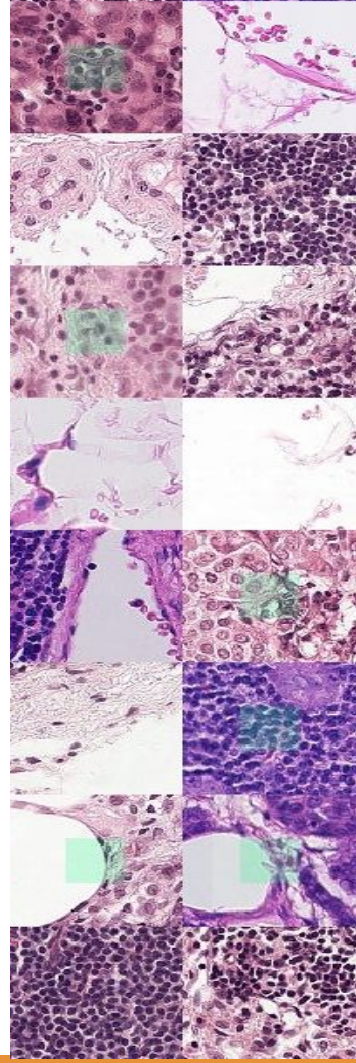


Encoding the Data

0 1 3
1 5 9
0 3 4
5 8 9
1 2 9
8 2 1
4 4 4
2 1 1
4 7 8
8 8 1
1 9 9
9 1 2

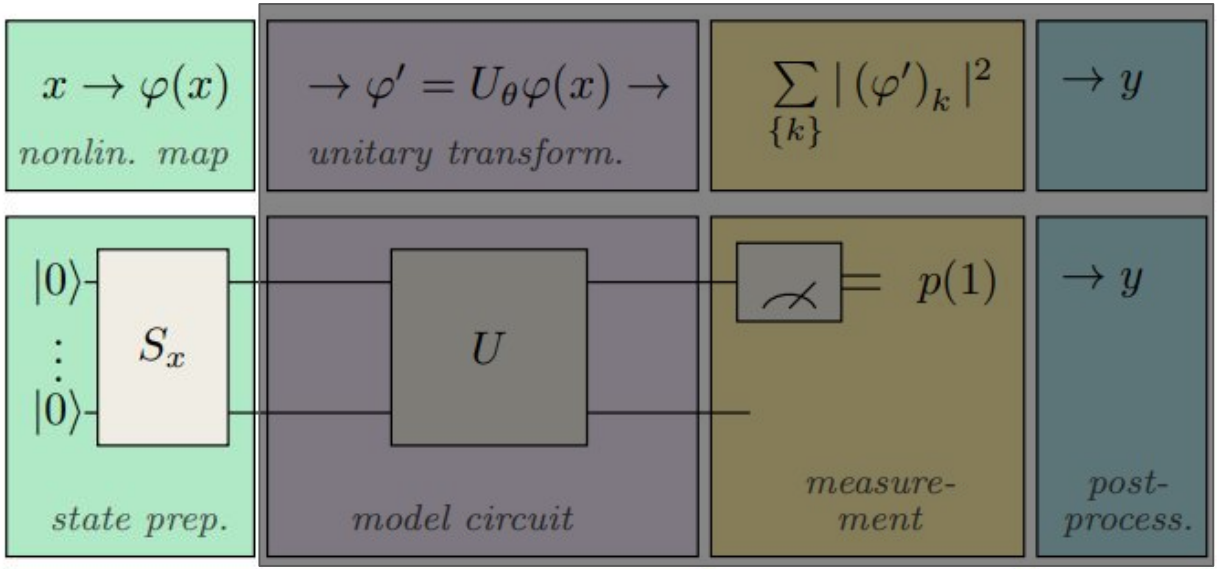


$$q_\phi(\mathbf{z}|\mathbf{x})$$

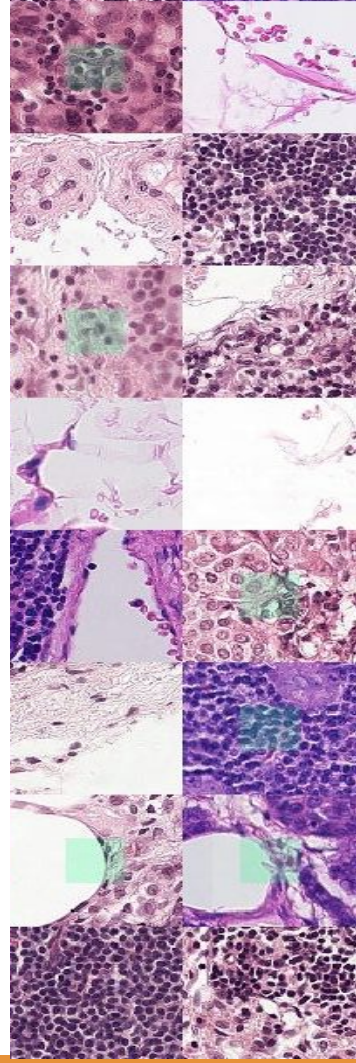


Encoding the Data

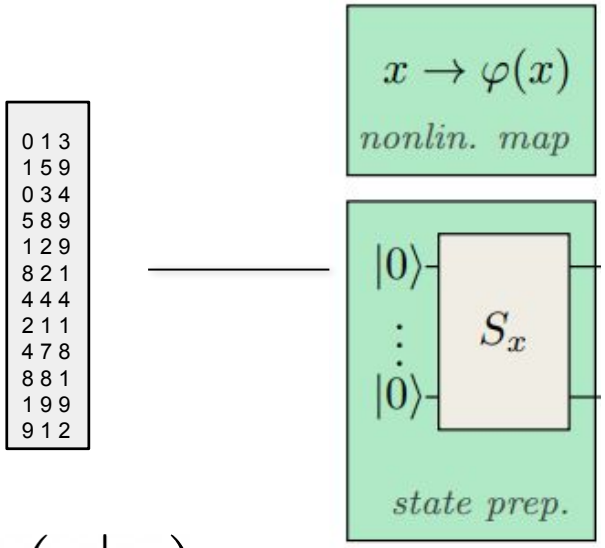
0 1 3
1 5 9
0 3 4
5 8 9
1 2 9
8 2 1
4 4 4
2 1 1
4 7 8
8 8 1
1 9 9
9 1 2



$$q_\phi(\mathbf{z}|\mathbf{x})$$



Encoding the Data



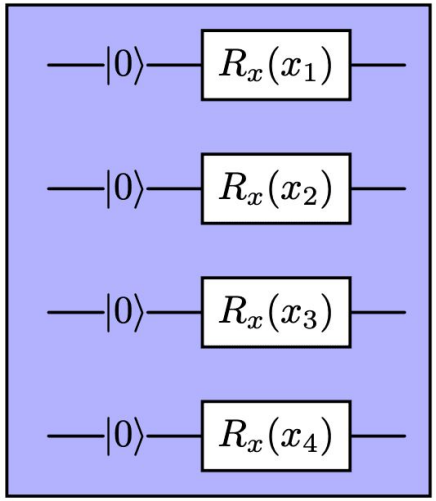
$q_\phi(\mathbf{z}|\mathbf{x})$

1. Angle Encoding
2. Amplitude Encoding

Encoding the Data

0 1 3
1 5 9
0 3 4
5 8 9
1 2 9
8 2 1
4 4 4
2 1 1
4 7 8
8 8 1
1 9 9
9 1 2

$q_\phi(\mathbf{z}|\mathbf{x})$



(a) Angle Encoding.

Angle Encoding

a. For each datapoint apply a rotation gate $R_P(\mathbf{x}_0)$
b. If we rotate around the Pauli-Y axis $R_Y(\mathbf{x}_0)$ we keep the state vector in the real plane

$$R_Y(\theta) = \begin{pmatrix} \cos(\theta/2) & -\sin(\theta/2) \\ \sin(\theta/2) & \cos(\theta/2) \end{pmatrix}$$

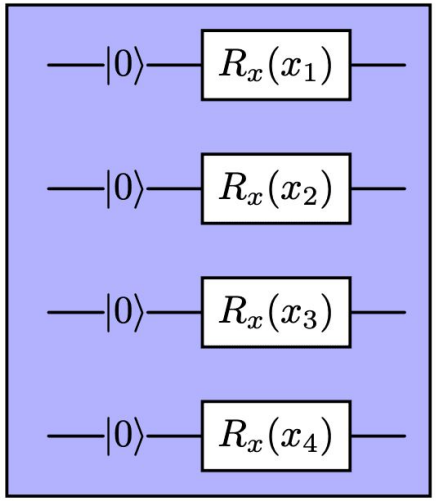
c. Requires *N* qubits

SURF

Encoding the Data

0 1 3
1 5 9
0 3 4
5 8 9
1 2 9
8 2 1
4 4 4
2 1 1
4 7 8
8 8 1
1 9 9
9 1 2

$$q_\phi(\mathbf{z}|\mathbf{x})$$



(a) Angle Encoding.

Angle Encoding

a. For each datapoint apply a rotation gate $R_P(\mathbf{x}_0)$

b. If we rotate around the Pauli-Z axis $R_Y(\mathbf{x}_0)$ we keep the state vector in the real plane

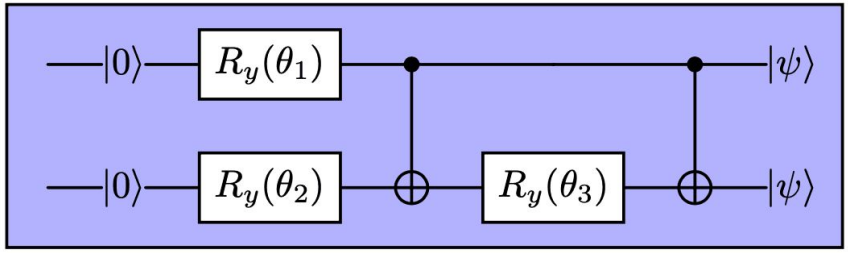
$$R_Z(\theta) = \begin{pmatrix} e^{-i\theta/2} & 0 \\ 0 & e^{i\theta/2} \end{pmatrix}$$

c. Can come in handy if data has phase relationships

SURF

Encoding the Data

$q_\phi(\mathbf{z}|\mathbf{x})$



(b) Amplitude Encoding.

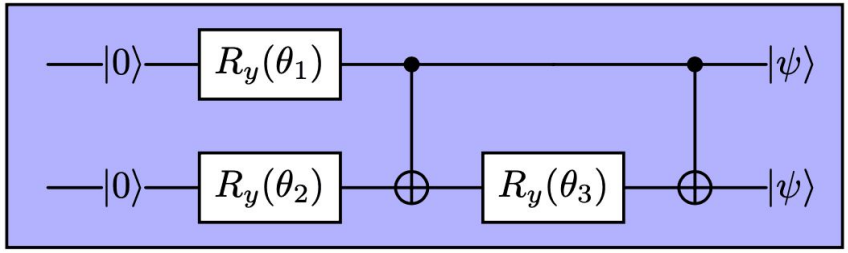
Amplitude Encoding

1. Encode the data vector into the amplitudes of the quantum state
2. Map the data to an N-dimensional Hilbert Space

3. State preparation has a non-negligible cost (on NISQ can negate benefit of qubit compression)
4. Encoded information is not directly accessible

SURF

Encoding the Data



(b) Amplitude Encoding.

Amplitude Encoding

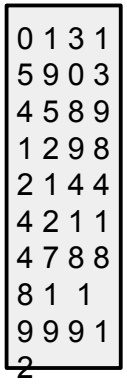
1. Encode the data vector into the amplitudes of the quantum state
2. Map the data to an N-dimensional Hilbert Space

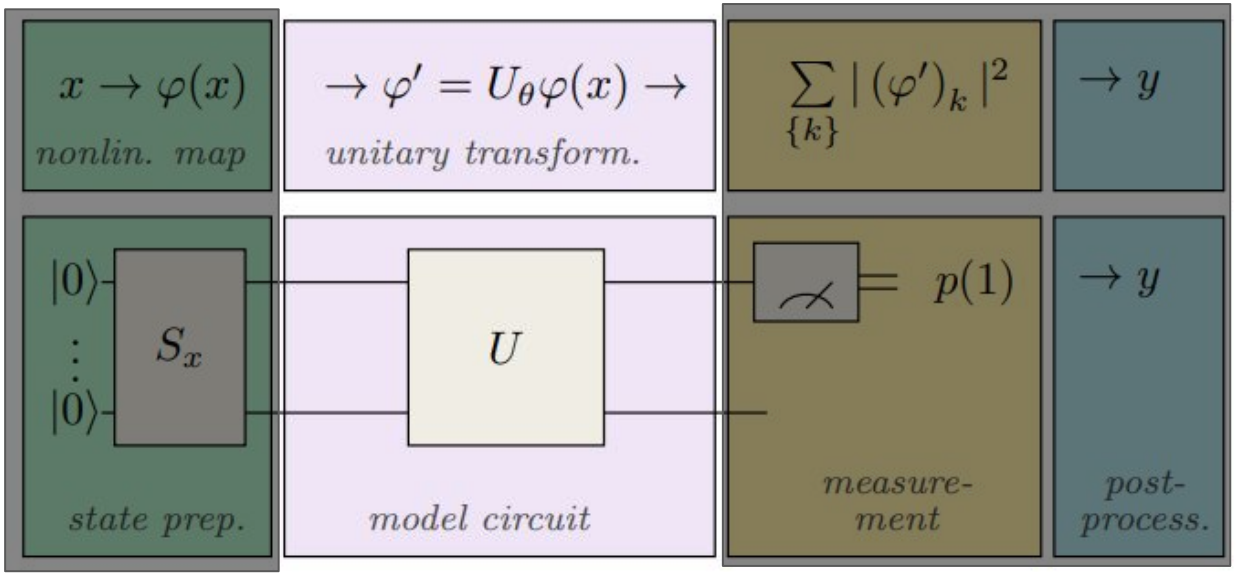
$$|\psi(\vec{x})\rangle = \frac{1}{||\vec{x}||} \sum_{i=1}^{N} x_i |i-1\rangle$$

3. Qubits needed $\lceil \log_2 N \rceil$

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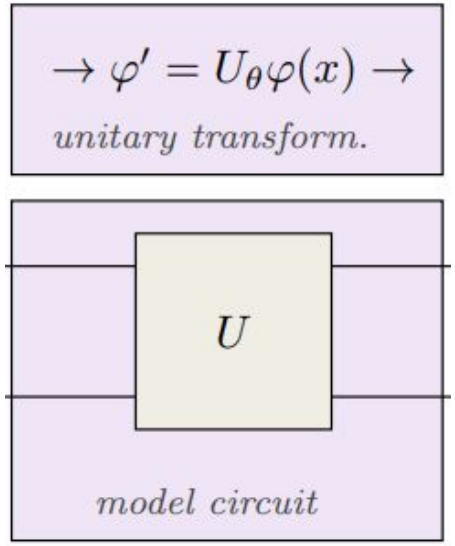
Training on the Data





$$q_\phi(\mathbf{z}|\mathbf{x})$$

Training on the Data



Two Types of Entangling for Training

1. Basic Entangling
2. Strongly Entangling

Apply a rotation gate with trainable parameter to each qubit

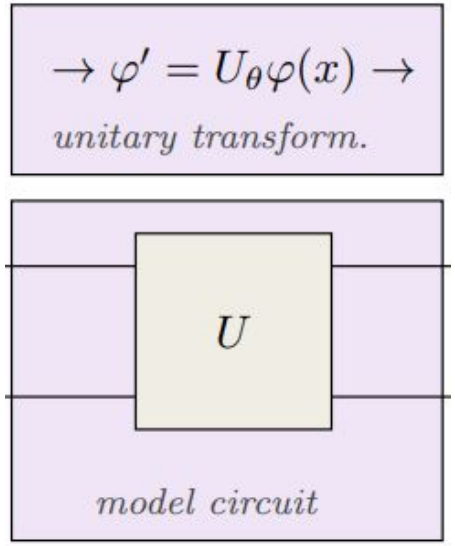
Apply a CNOT gate between adjacent qubits

VS

Apply a CNOT gate between every qubit and every other qubit

SURF

Training on the Data



Two Types of Entangling for Training

1. Basic Entangling
2. Strongly Entangling

Apply a rotation gate with trainable parameter to each qubit

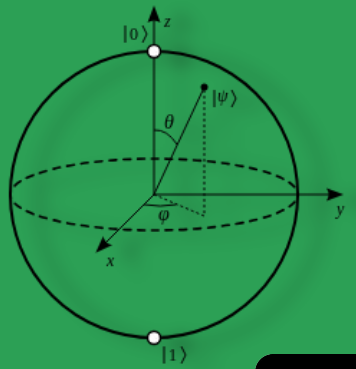
Apply a CNOT gate between adjacent qubits

VS

Apply a CNOT gate between every qubit and every other qubit

SURF

Results



Parameters

1. **Classical ML**
   Encoder - Classifier Model on PCAM

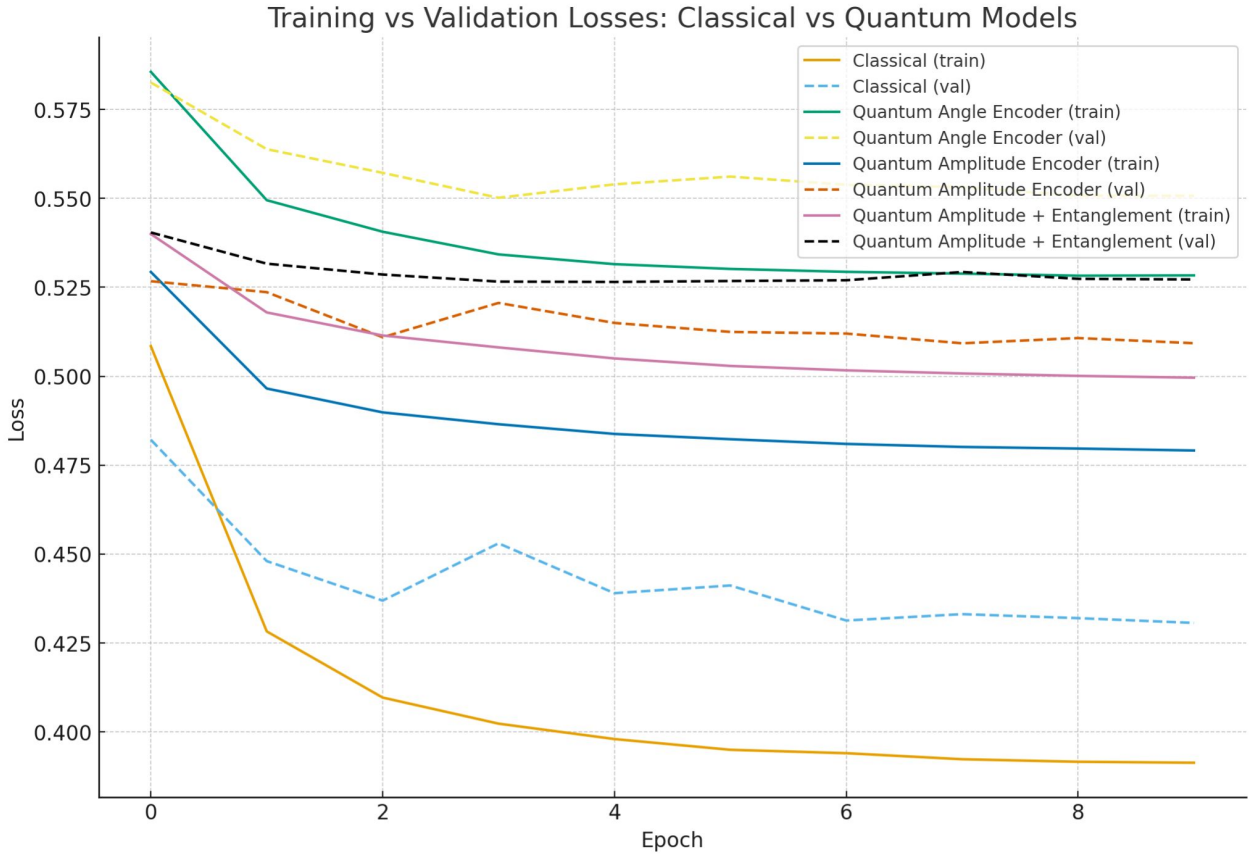
2. **Hybrid QML**
   Encoder - VQC Model

1. Map to a small latent dimension, vary classical model hyperparameters
2. AmplitudeEmbedding **vs** AngleEmbedding
3. Experiment with Strongly and Basic Entangler Layers

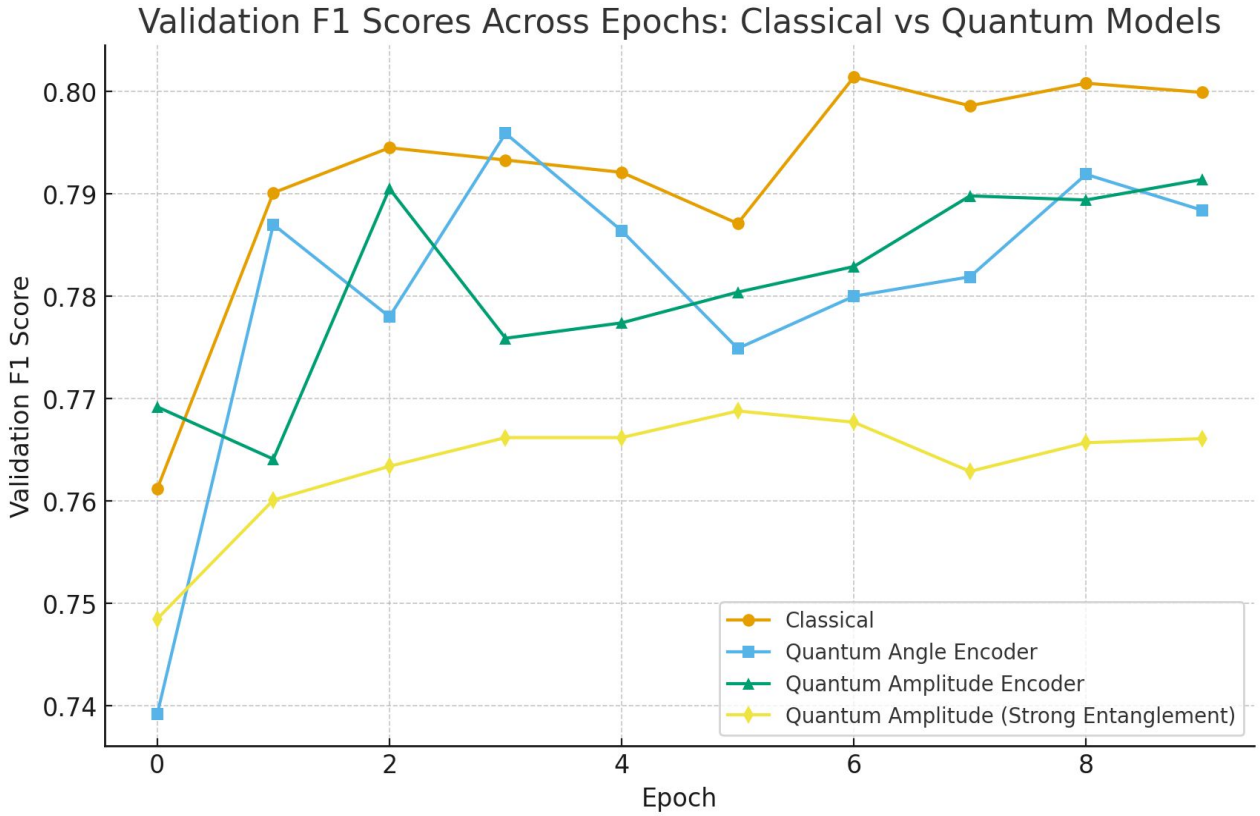
Make it hard for the classical model to extract good features
Reduce (or increase) the number of trainable quantum parameters

SURF

Results: PCAM

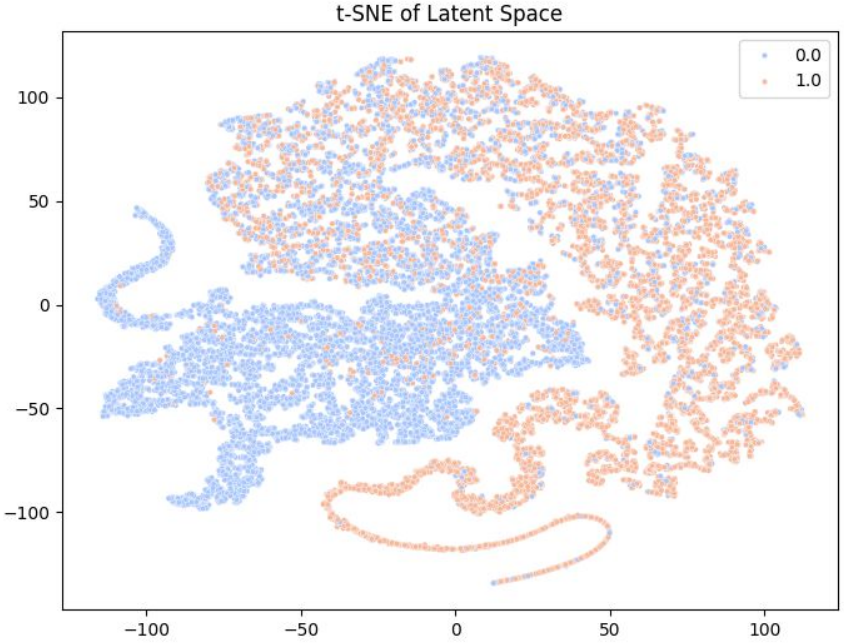


Results: PCAM

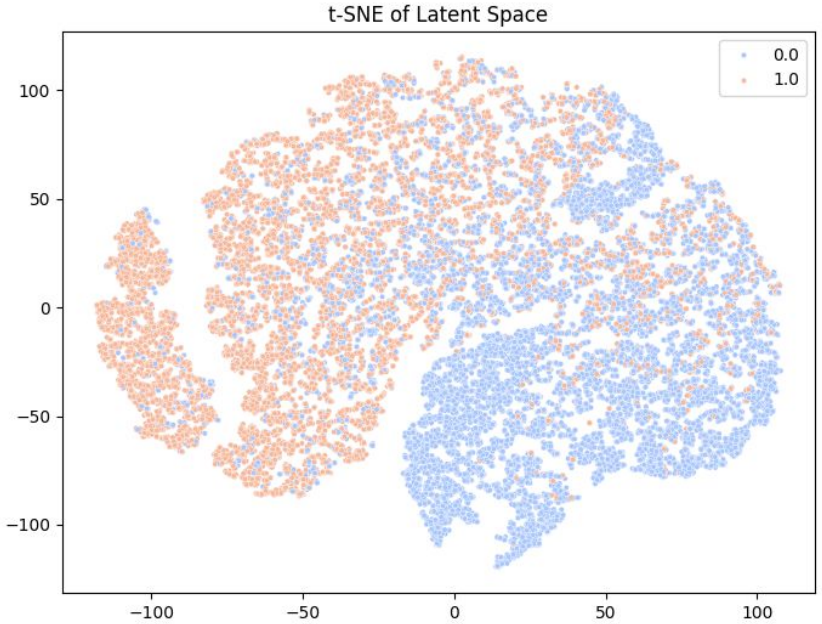


Results: PCAM

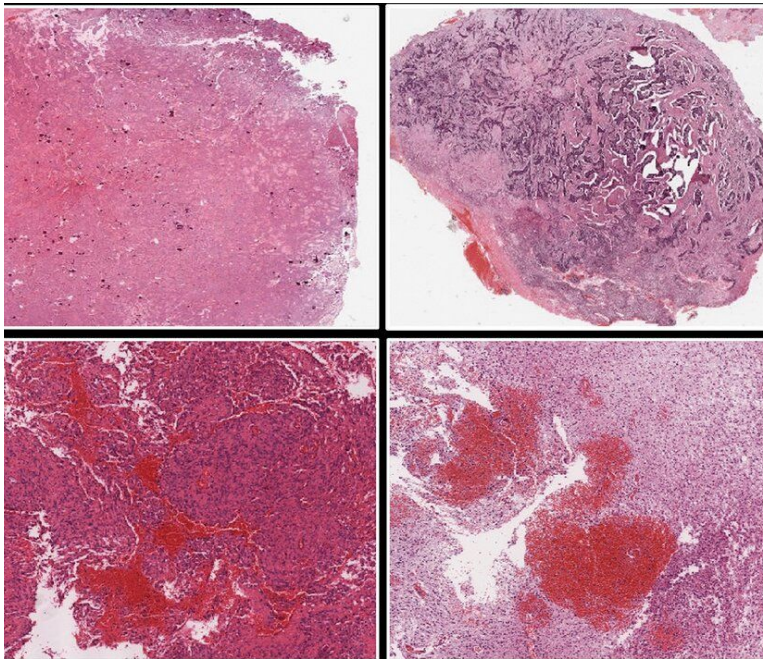
Classical Classifier



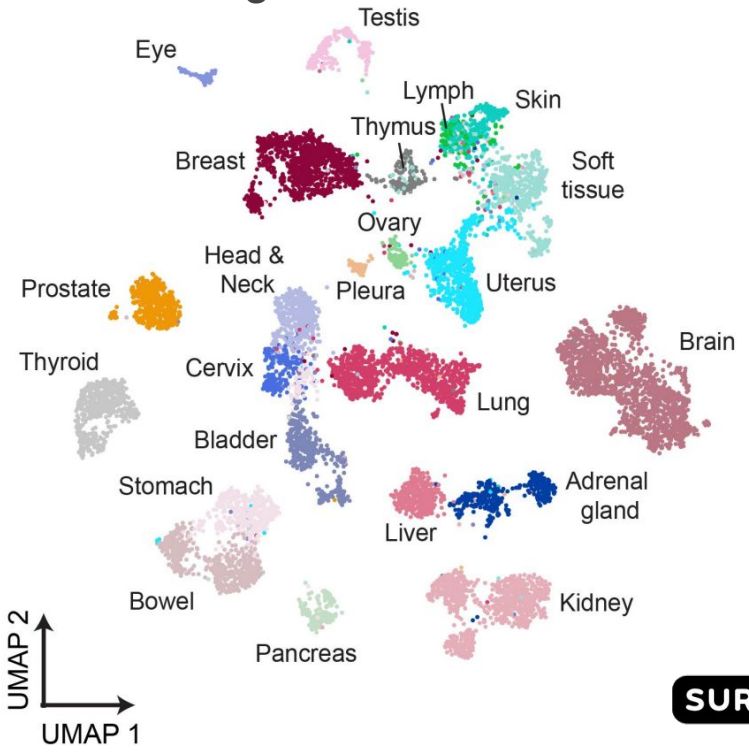
Amplitude Embedding



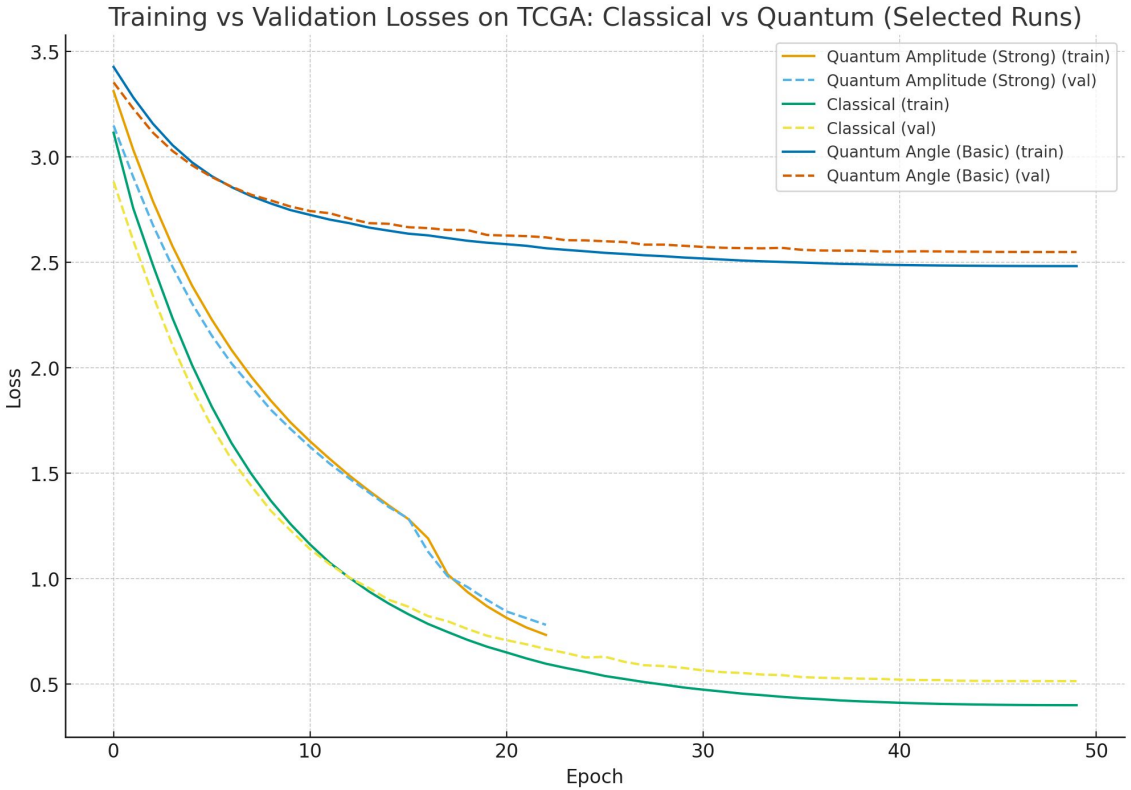
Results: TCGA



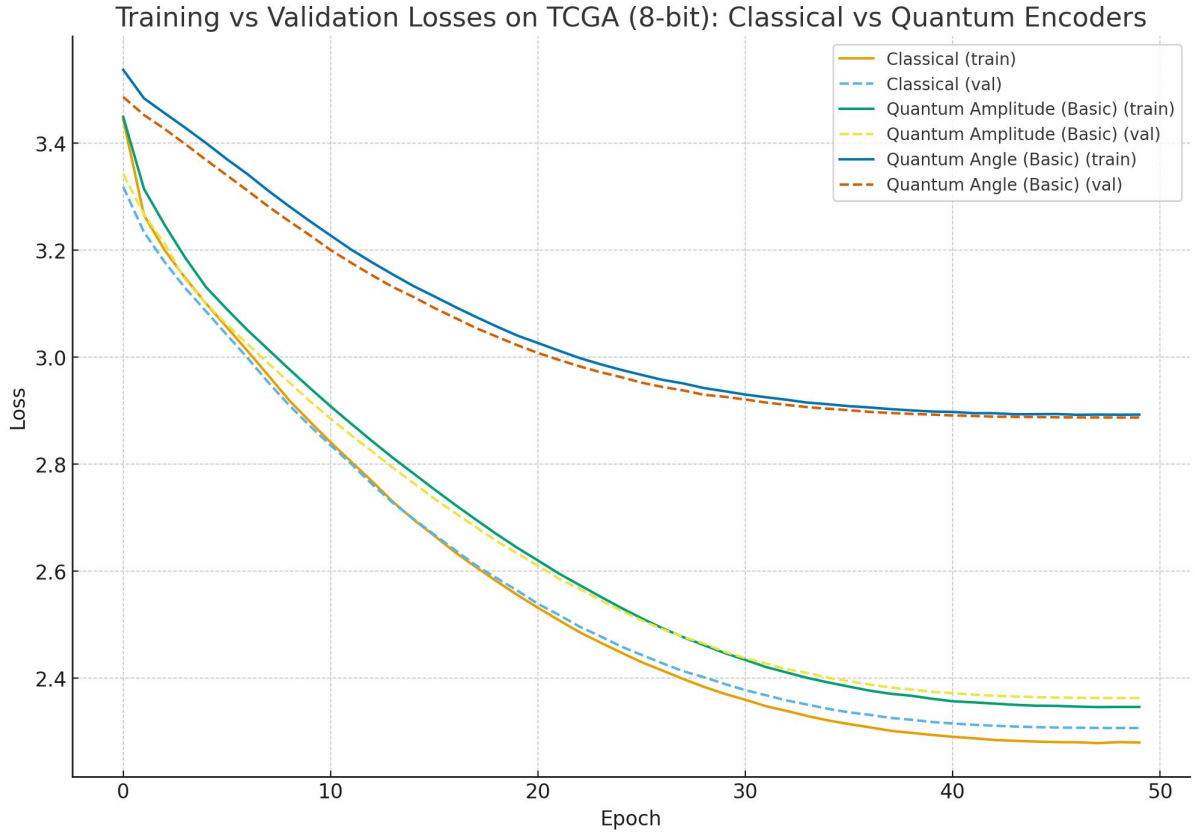
TITAN Classical Encoder Embeddings



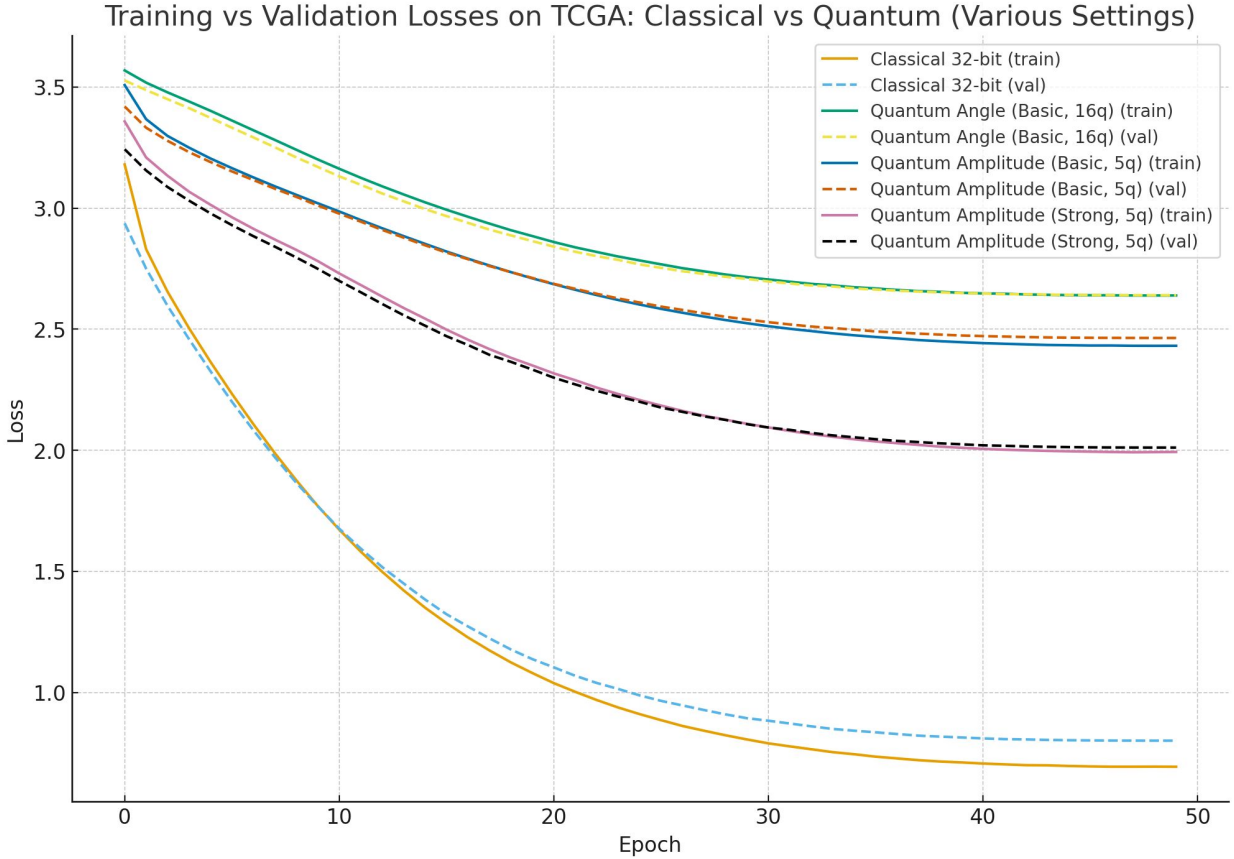
Results: TCGA



Results: TCGA

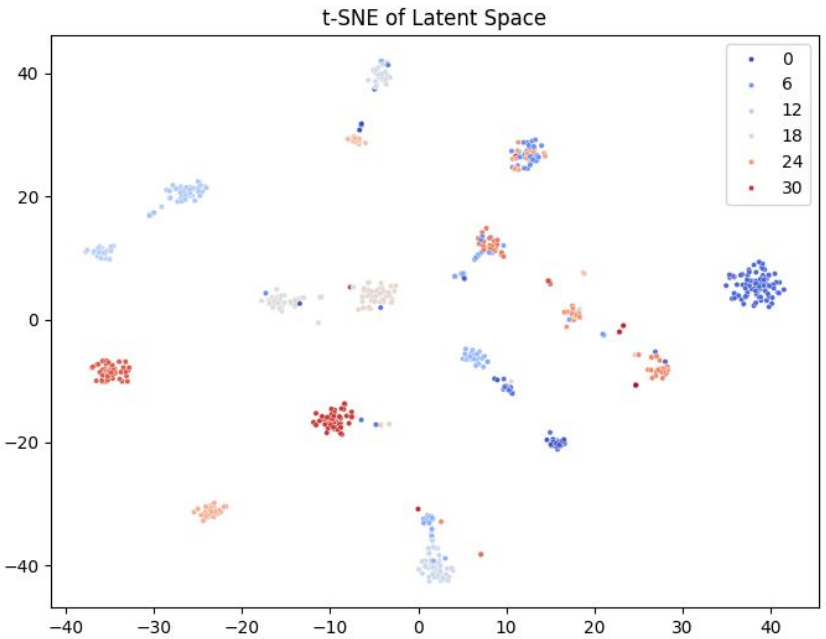


Results: TCGA

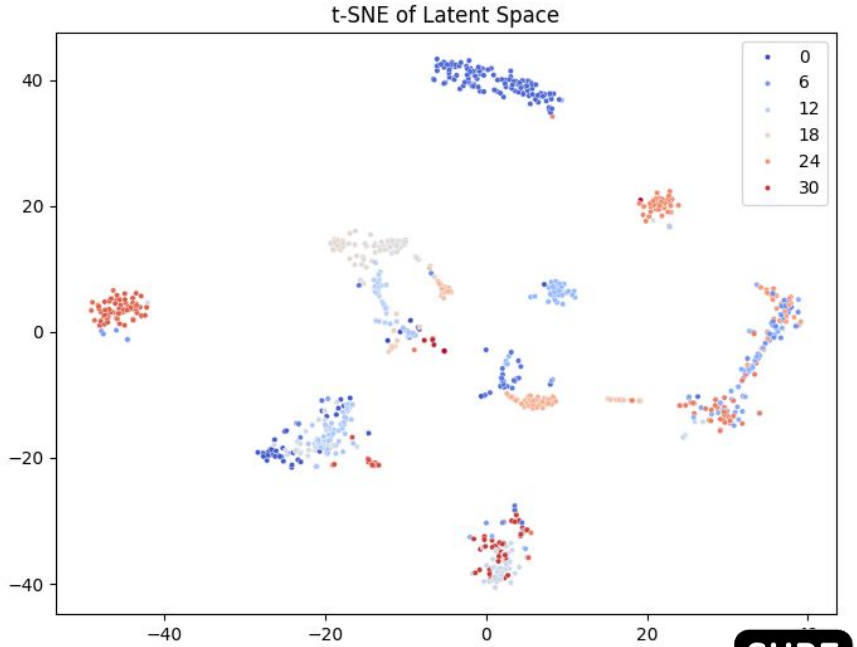


Results: TCGA

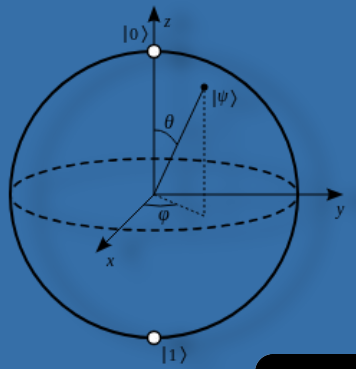
Classical Classifier



Amplitude Embedding



Importance of Inductive Bias



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Core Concept

1. **Linear Models**
   Assume relationship between features and outcome is a hyperplane

2. **CNNs**
   Bias for locality and translation invariance

3. **RNNs**
   Bias for sequences and features through time

4. **Self-Attention**
   Bias for sequences but permutation invariant

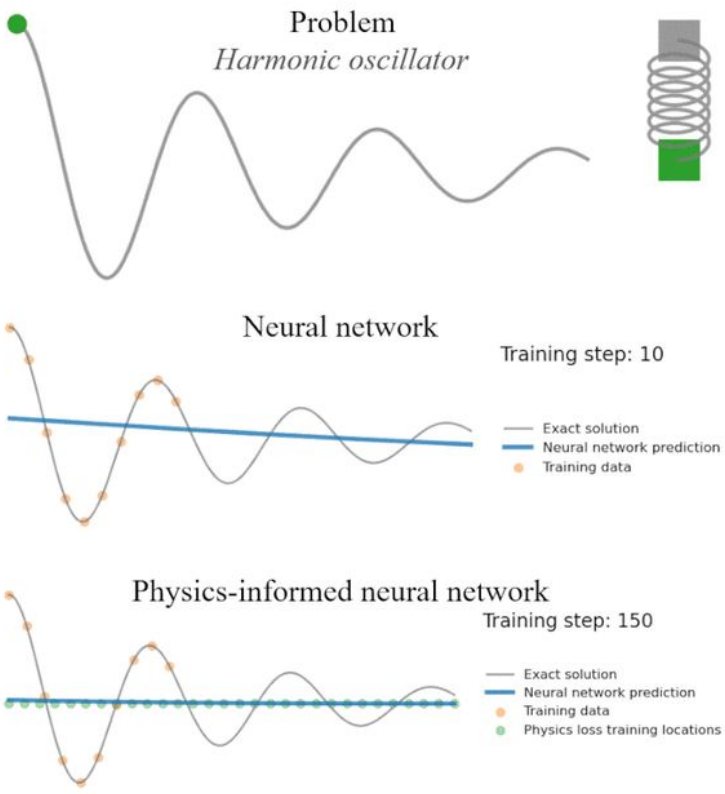
Core Concept

1. **Linear Models**
   Assume relationship between features a outcome is a hyperplane

2. **CNNs**
   Bias for locality and translation invariance

3. **RNNs**
   Bias for sequences and features through time

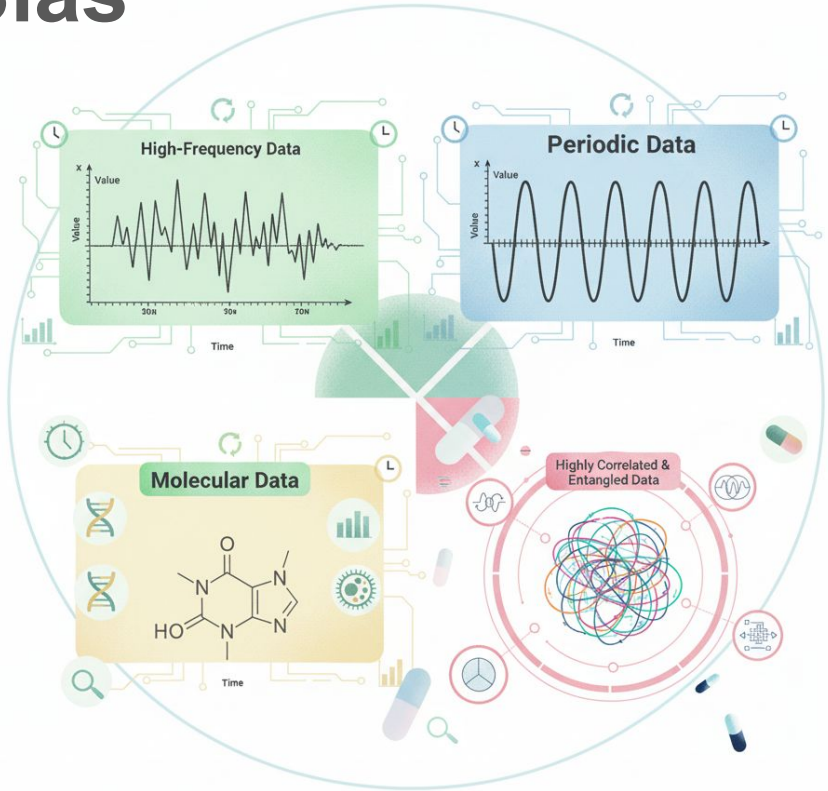
4. **Self-Attention**
   Bias for sequences but permutation invariant



Quantum Inductive Bias

1. **Periodicity & Oscillatory**
   VQC is fundamentally a sequence of rotation gates (Rx, Ry, Rz), output is a Fourier series

2. **Entangled Correlations**
   Model non-local correlations



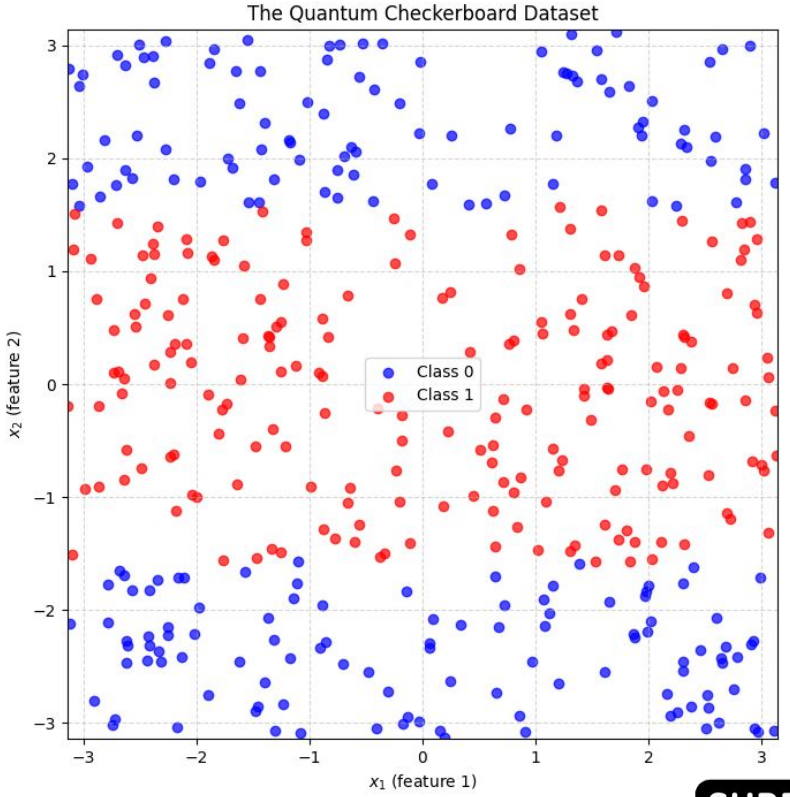
The Checkerboard Problem

1. **Linear Model**
   Will try to separate the data

$$\hat{y} = \sum_{i=1}^{m} w_i x_i + b$$

2. **Angle Encoding + BasicEntangling**
   Inherently non-linear trigonometric mapping

$$R_Y(\theta) = \begin{pmatrix} \cos(\theta/2) & -\sin(\theta/2) \\ \sin(\theta/2) & \cos(\theta/2) \end{pmatrix}$$



SURF

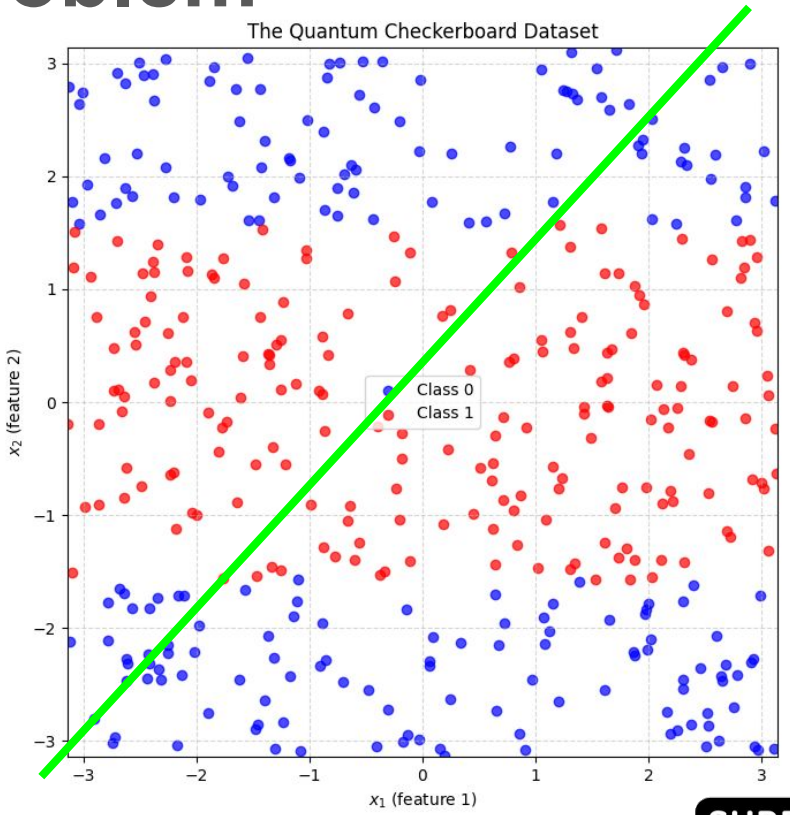
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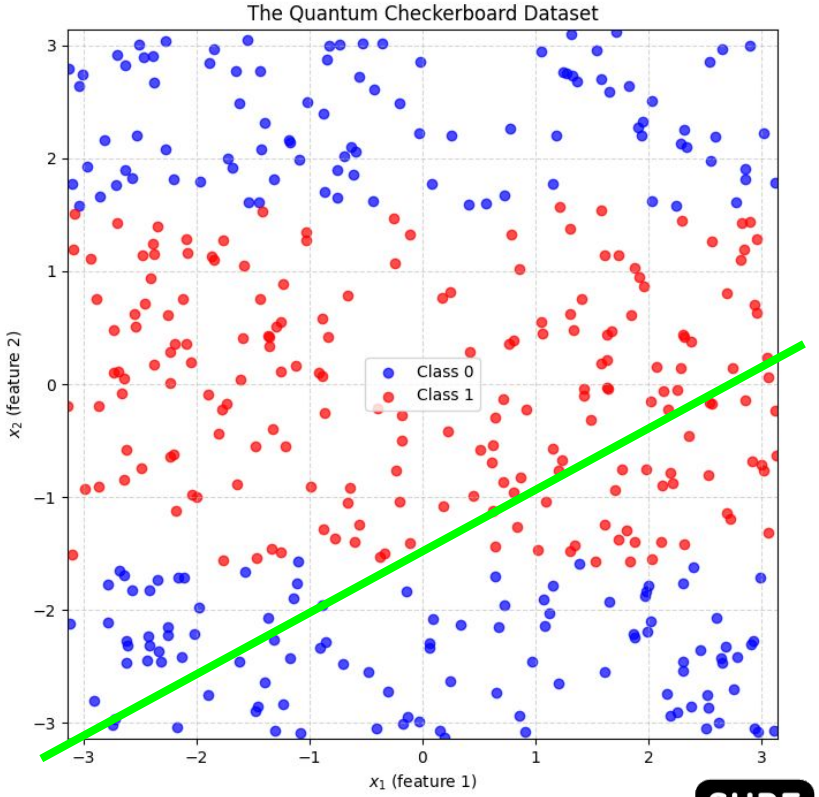
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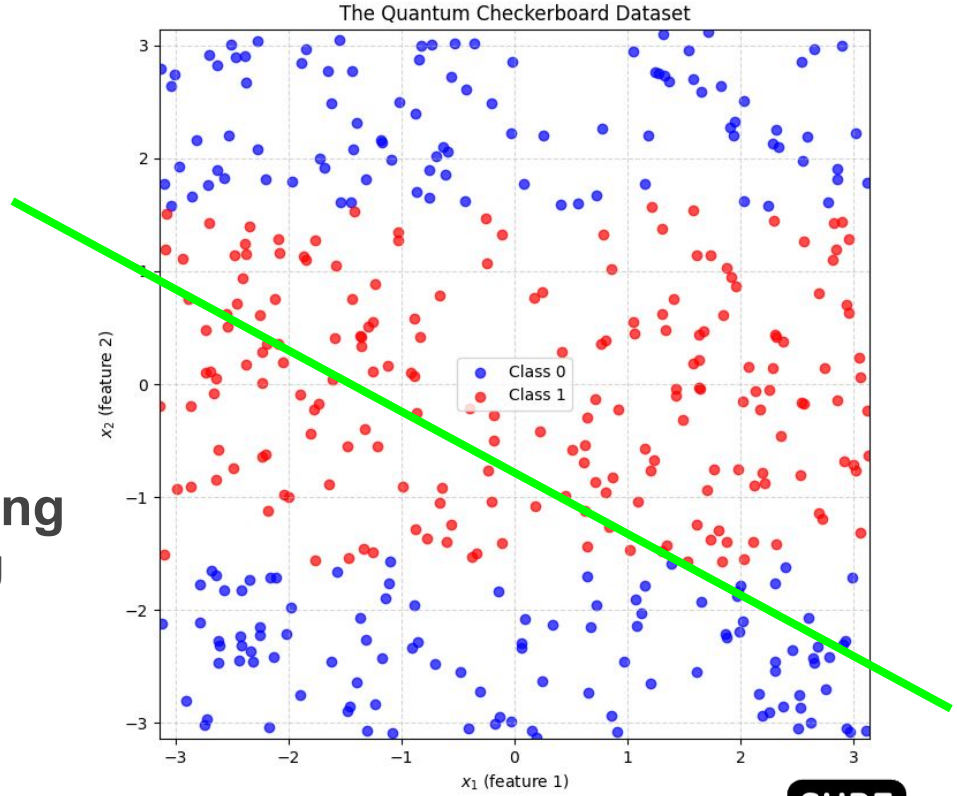
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SURF

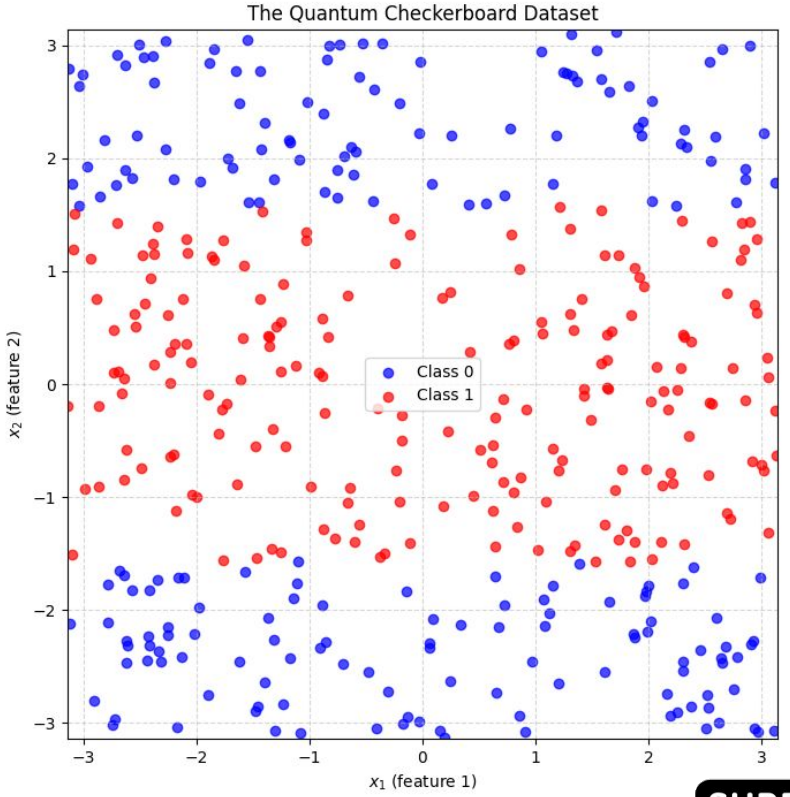
The Checkerboard Problem

```
(venv) [bryanc@gcn142 src]$ python checkerboard_demo.py
Data generation complete.
Data visualization saved to checkerboard_plot.png

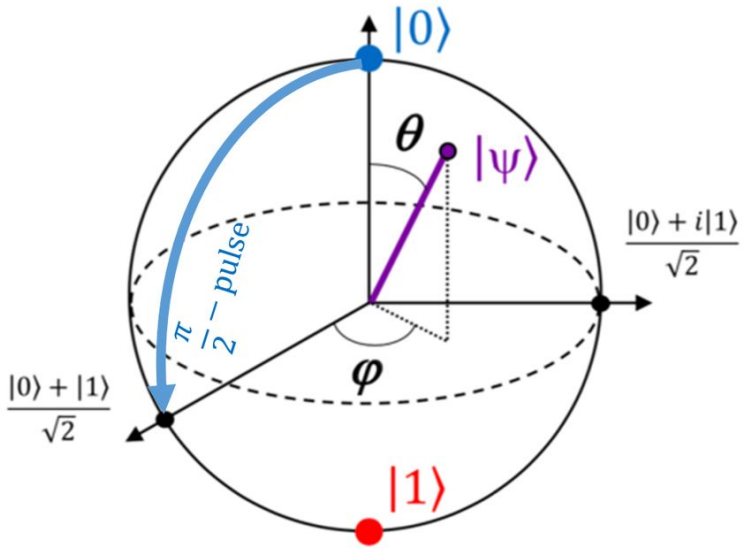
--- Training ClassicalClassifier ---
Epoch  5/20 | Loss: 0.6979 | Accuracy: 0.4825
Epoch 10/20 | Loss: 0.7007 | Accuracy: 0.3850
Epoch 15/20 | Loss: 0.7061 | Accuracy: 0.4800
Epoch 20/20 | Loss: 0.6972 | Accuracy: 0.5275
Training finished.
Final Accuracy for ClassicalClassifier: 0.5275

--- Training QuantumClassifier ---
Epoch  5/20 | Loss: 0.4385 | Accuracy: 0.9925
Epoch 10/20 | Loss: 0.4410 | Accuracy: 0.9900
Epoch 15/20 | Loss: 0.4382 | Accuracy: 0.9925
Epoch 20/20 | Loss: 0.4375 | Accuracy: 0.9925
Training finished.
Final Accuracy for QuantumClassifier: 0.9925

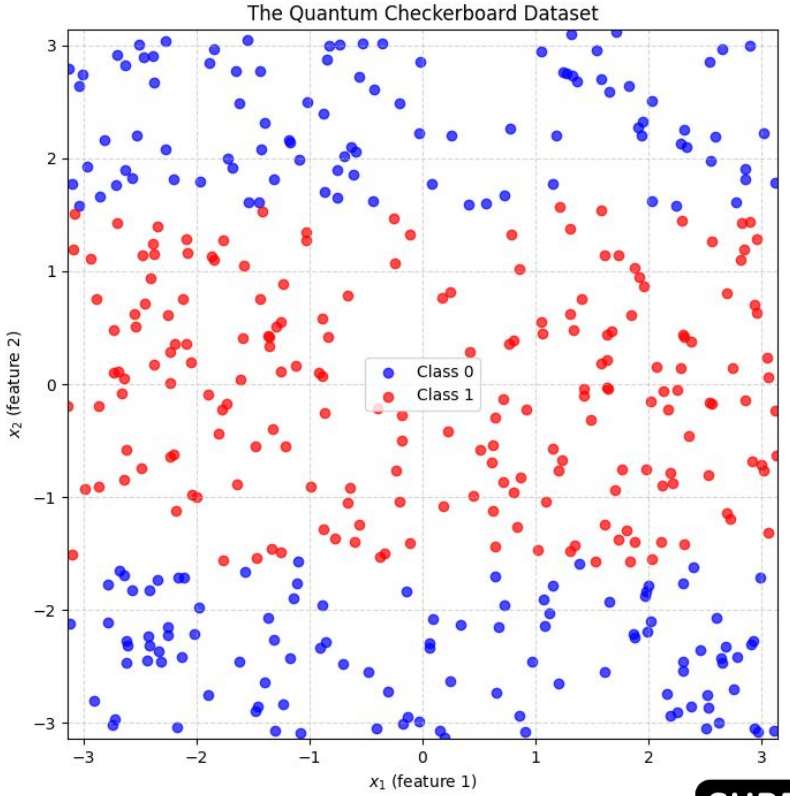
--- Comparison ---
Classical Model Final Accuracy: 0.5275
Quantum Model Final Accuracy:   0.9925
```



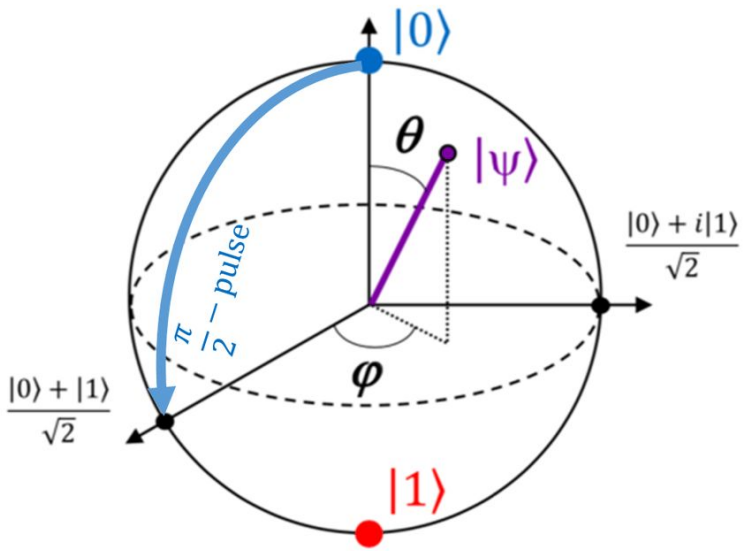
The Checkerboard Problem



$$|\psi\rangle = \cos\frac{\theta}{2}|0\rangle + e^{i\varphi}\sin\frac{\theta}{2}|1\rangle$$



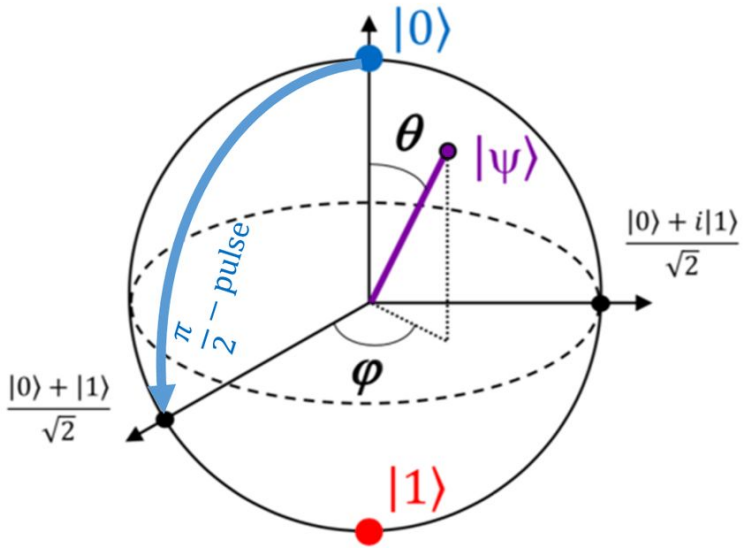
The Checkerboard Problem



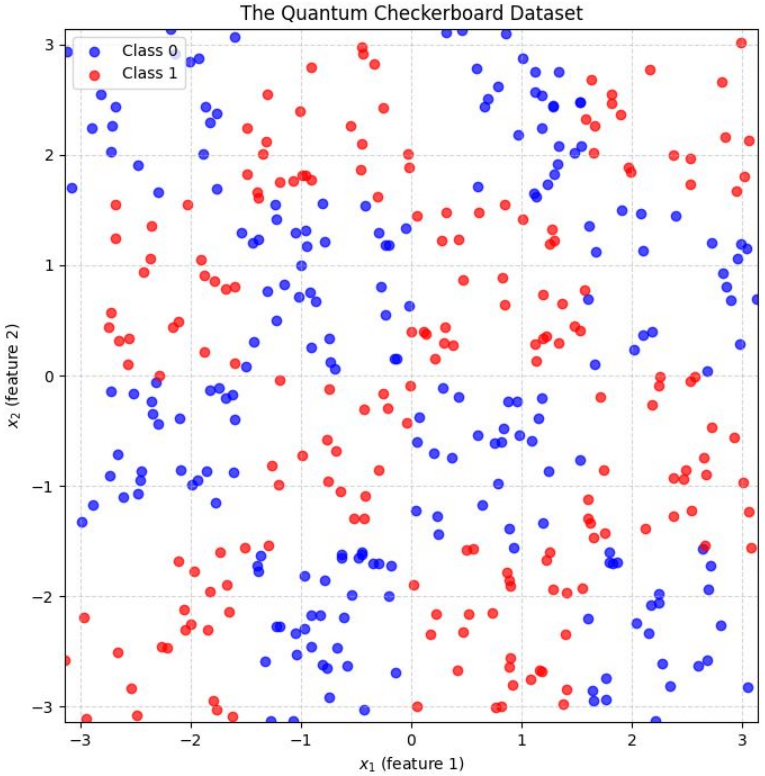
$$|\psi\rangle = cos\frac{\theta}{2}|0\rangle + e^{i\varphi}sin\frac{\theta}{2}|1\rangle$$

1. Rotate state vector around X, Y, Z axis of the bloch sphere
2. **Before** entanglement both qubits are independent
3. **After** entanglement data has been encoded to combined 4-dimensional complex vector space

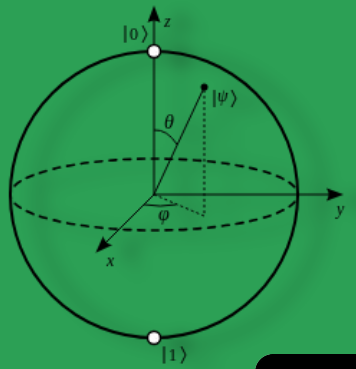
Can we make it work for the checkerboard?



$$|\psi\rangle = \cos\frac{\theta}{2}|0\rangle + e^{i\varphi}\sin\frac{\theta}{2}|1\rangle$$



Lessons from literature



SURF

Lessons from Literature

1. **Image (hybrid) classification models**
   Achieves parity but no clear advantage

2. **Inductive bias proliferates**
   Advantage appears when data fits quantum friendly formats

3. **When data is too “easy”**

   No advantage, classical reigns supreme

4. **No broad advantage**

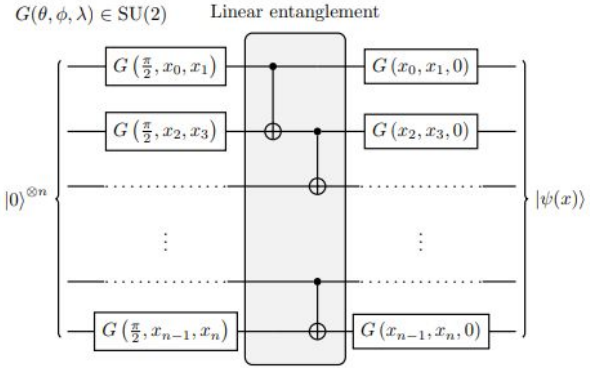
   Current advantages are incremental

https://quantum.cern/sites/default/files/2023-05/Quantum%20Algorithms%20for%20anomaly%20detection%20at%20the%20LHC.pdf

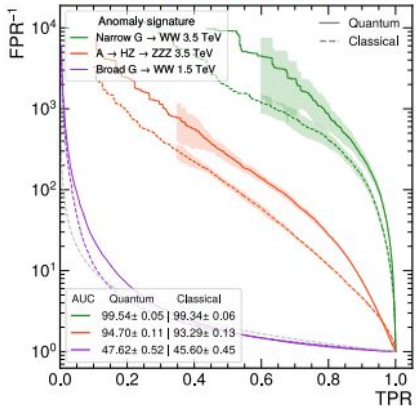
Kernel-based quantum anomaly detection

Unsupervised quantum kernel machine $\mathrm{K}_{ij} = |\langle 0|U^{\dagger}(\vec{x}_i)U(\vec{x}_j)|0\rangle|^2$

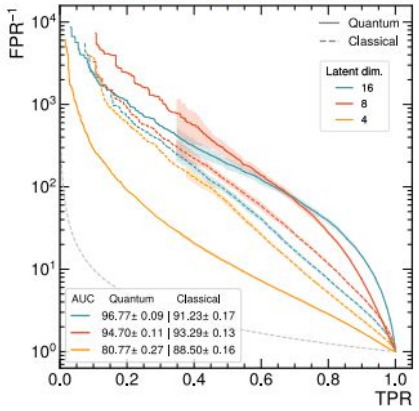
Designed data encoding circuit



Different BSM scenarios



Different qubit number

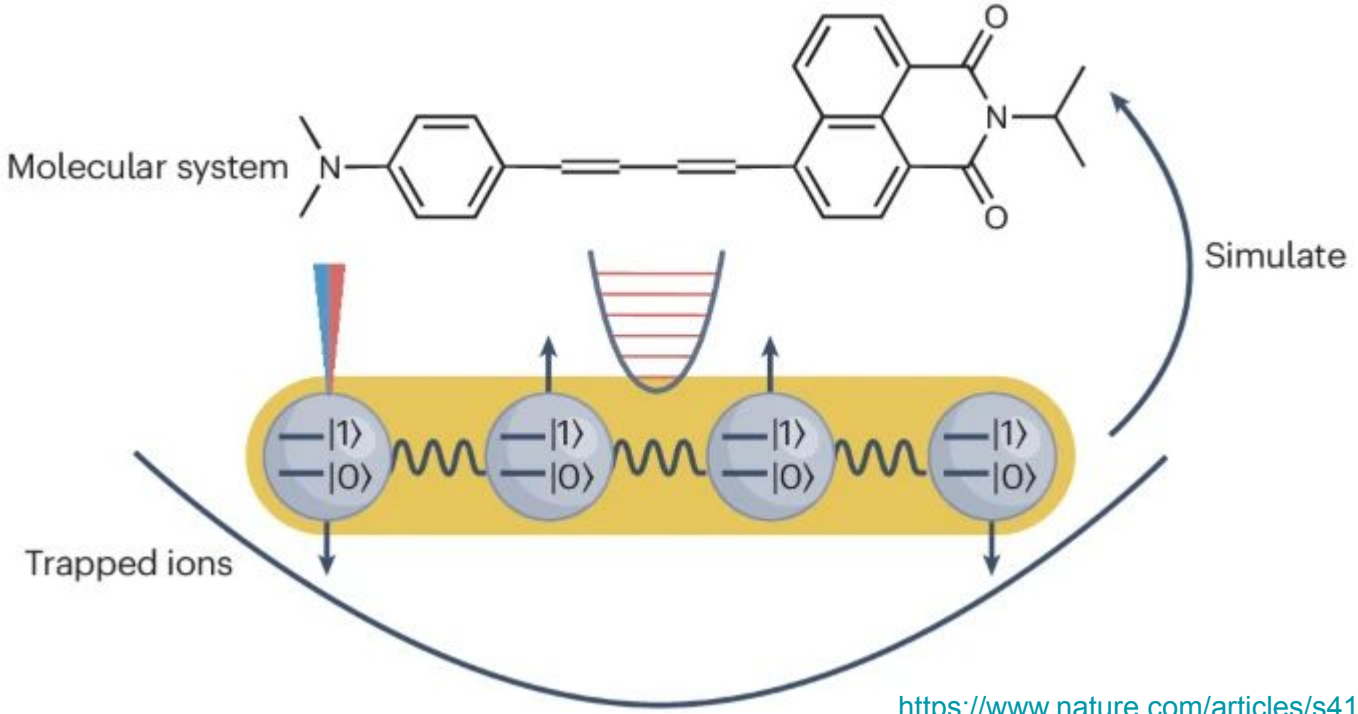


[K.A. Wozniak*, VB*, E. Puljak*, et al., arXiv: 2301.10780]

Instance of significant and consistent quantum performance advantage!

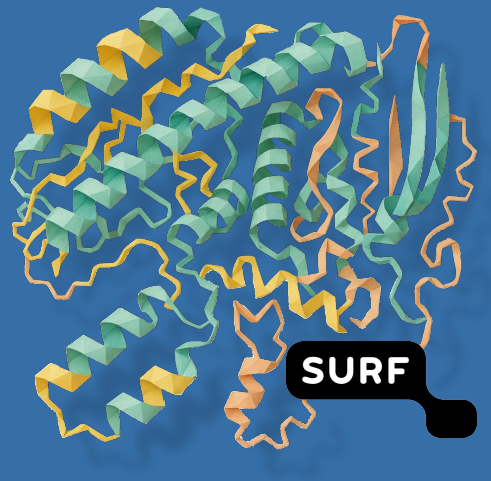
Very exciting and first of its kind result (HEP + Anomaly detection)!

Simulating Compounds

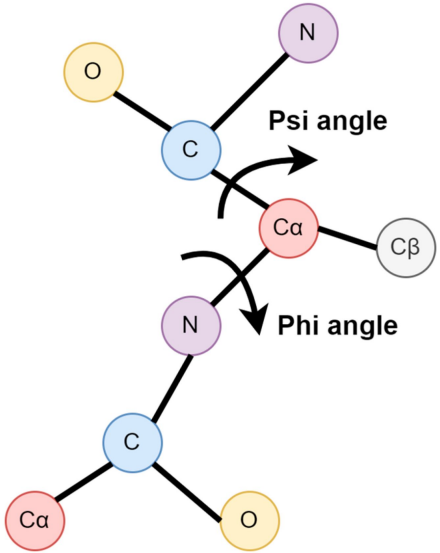


https://www.nature.com/articles/s41570-024-00595-1

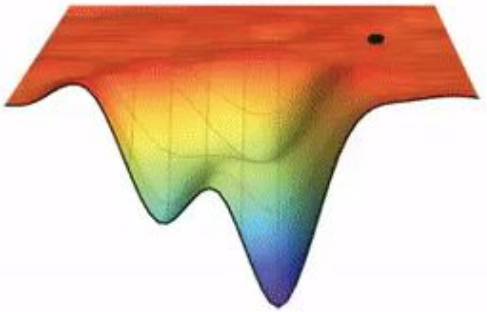
What can we do?



Torsion Angles







AlphaFold-2: Invariant Point Attention

From **representations** to concrete **3D coordinates**

Parameterize torsion angles for each amino-acid

Don’t learn rotations or translations in space

Structure module

© 2020 DeepMind Technologies Limited

- **End-to-end folding** instead of gradient descent
- Protein backbone = gas of 3-D rigid bodies (chain is learned!)

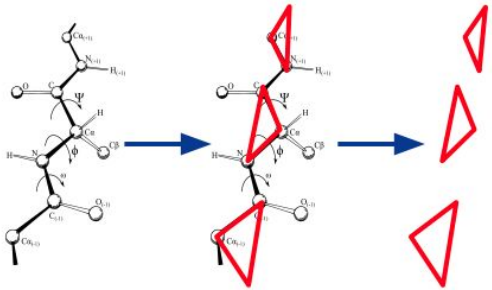
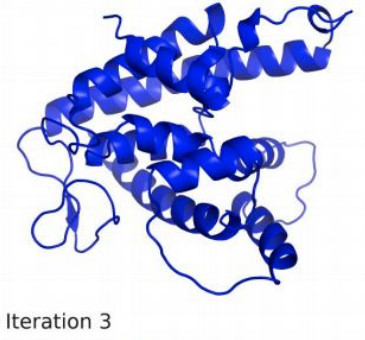
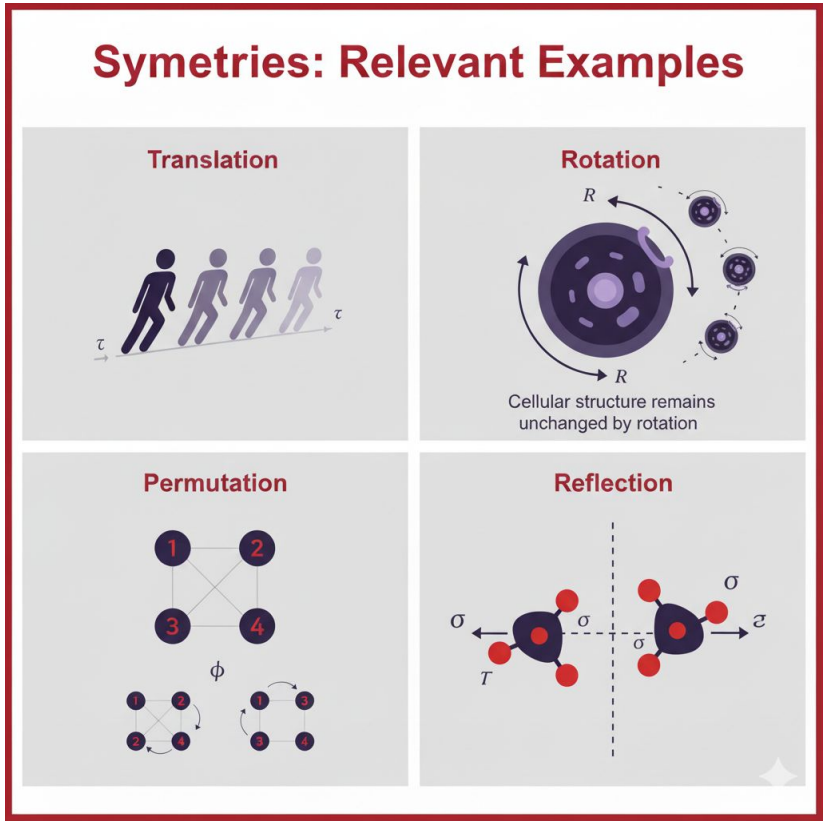


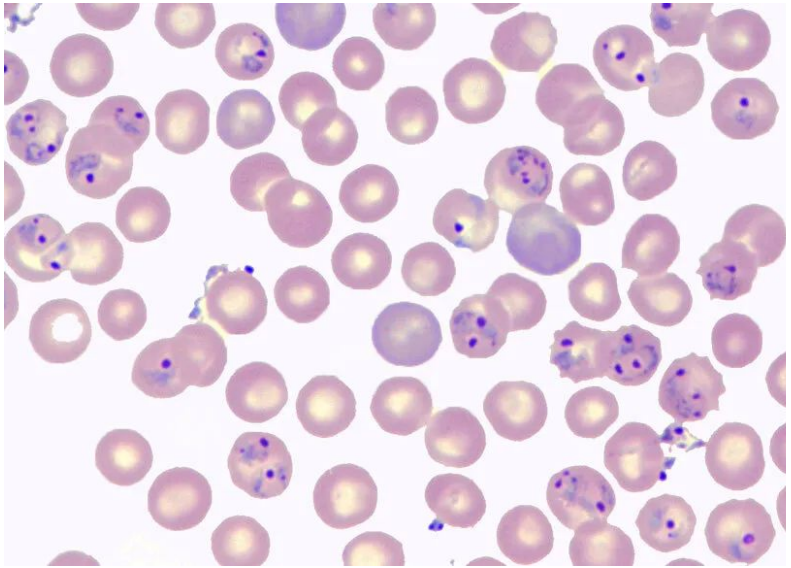
Image: Dcrjsr, vectorised Adam Rędzikowski (CC BY 3.0, Wikipedia)

- **3-D equivariant transformer architecture** updates the rigid bodies / backbone
  - Also builds the side chains



Medical Images: Invariances





Thanks!